

ARCOS Group

uc3m | Universidad **Carlos III** de Madrid

L3: Fundamentals of assembler programming (4) Computer Structure

Bachelor in Computer Science and Engineering

Bachelor in Applied Mathematics and Computing

Dual Bachelor in Computer Science and Engineering and Business Administration



Contents

- ▶ Basic concepts on assembly programming
- ▶ RISC-V 32 assembly language, memory model and data representation
- ▶ Instruction formats and addressing modes
- ▶ **Procedure calls and stack convention**
 - ▶ How do you call a function/subroutine?
 - ▶ Where is the return address stored in non-terminal routines?
 - ▶ What is the parameter passing convention?
 - ▶ What is the register use agreement?
 - ▶ What are the local variables like?

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Procedures and functions

```
int factorial(int x) {  
    int i;  
    int r=1;  
    for (i=1;i<=x;i++) {  
        r*=i;  
    }  
    return r;  
}  
...  
r1 = factorial(3) ;  
...
```

- ▶ A high-level function (procedure, method, etc.) is a subprogram that:
 - ▶ Receives input arguments or parameters
 - ▶ Performs a specific task when invoked
 - ▶ Returns some result

Functions in a high-level language

Steps in the execution of a function

```
int main() {  
    int z;  
    1 x=3;  
    z=factorial(x);  
    print_int(z);  
}
```

```
int factorial(int x) {  
    int i;  
    int r=1;  
    for (i=1;i<=x;i++) {  
        r*=i;  
    }  
    return r;  
}
```

1. Place the parameters in a place where they can be accessed by the function
2. Transfer the flow control to the function
3. Acquire storage resources needed for the function
4. Perform the desired task
5. Store the result where the calling function can access it
6. Return control to the point of origin

Functions in a high-level language

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2  
→  
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    }  
    return r;  
}
```

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}
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3



```
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        r*=i;  
    }  
    return r;  
}
```

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Steps in the execution of a function

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    int z;  
    x=3;  
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    print_int(z);  
}
```

Local variables

```
int factorial(int x) {  
    int i;  
    int r=1;  
    for (i=1;i<=x;i++) {  
        r*=i;  
    }  
    return r;  
}
```

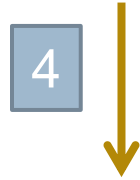
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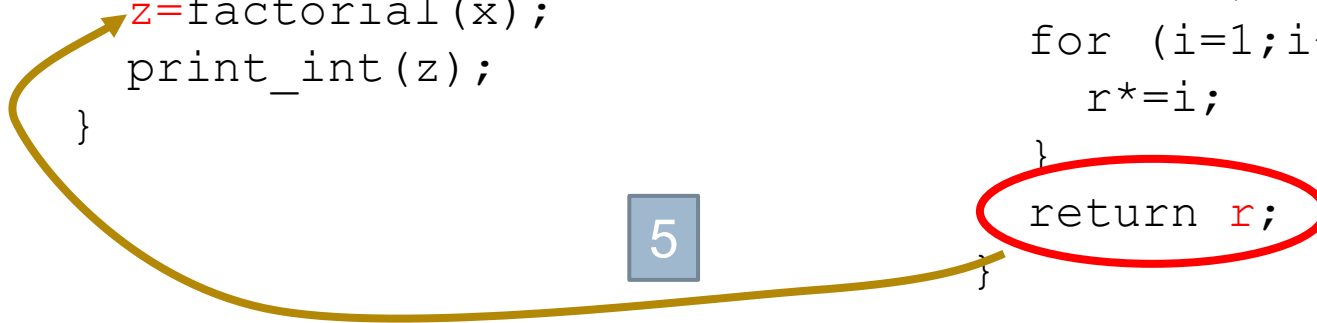
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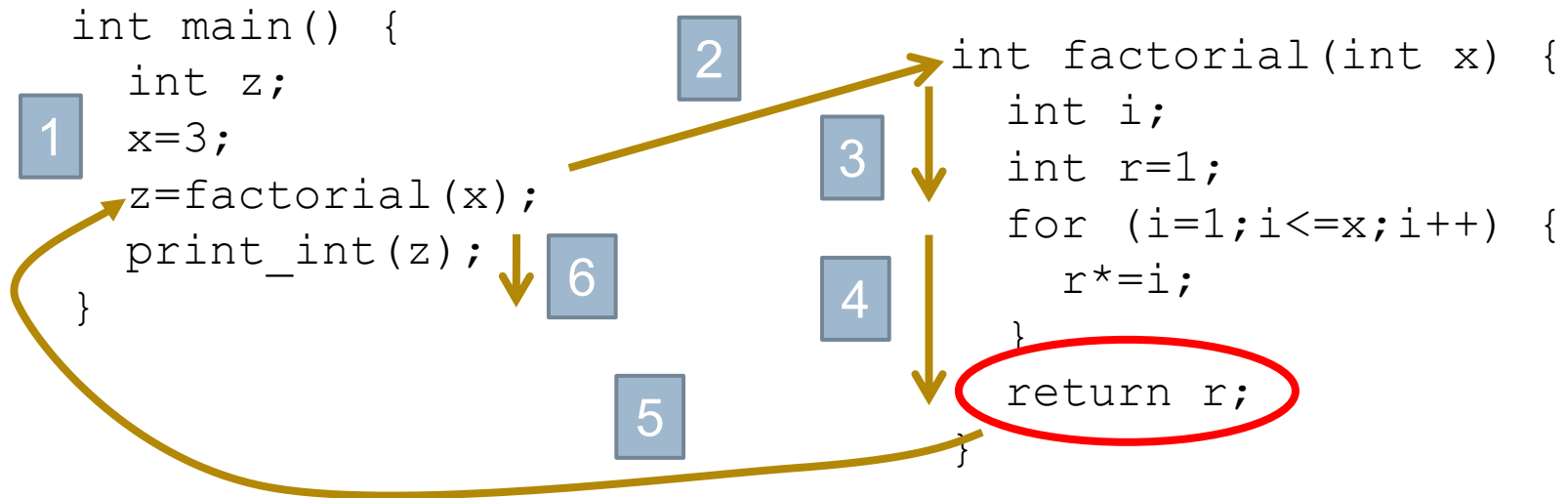
↓ 6

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    int i;  
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        r*=i;  
    }  
    return r;  
}
```

1. Place the parameters in a place where they can be accessed by the function
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Steps in the execution of a function de alto nivel

summary



1. Place the parameters in a place where they can be accessed by the function
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Procedures and functions

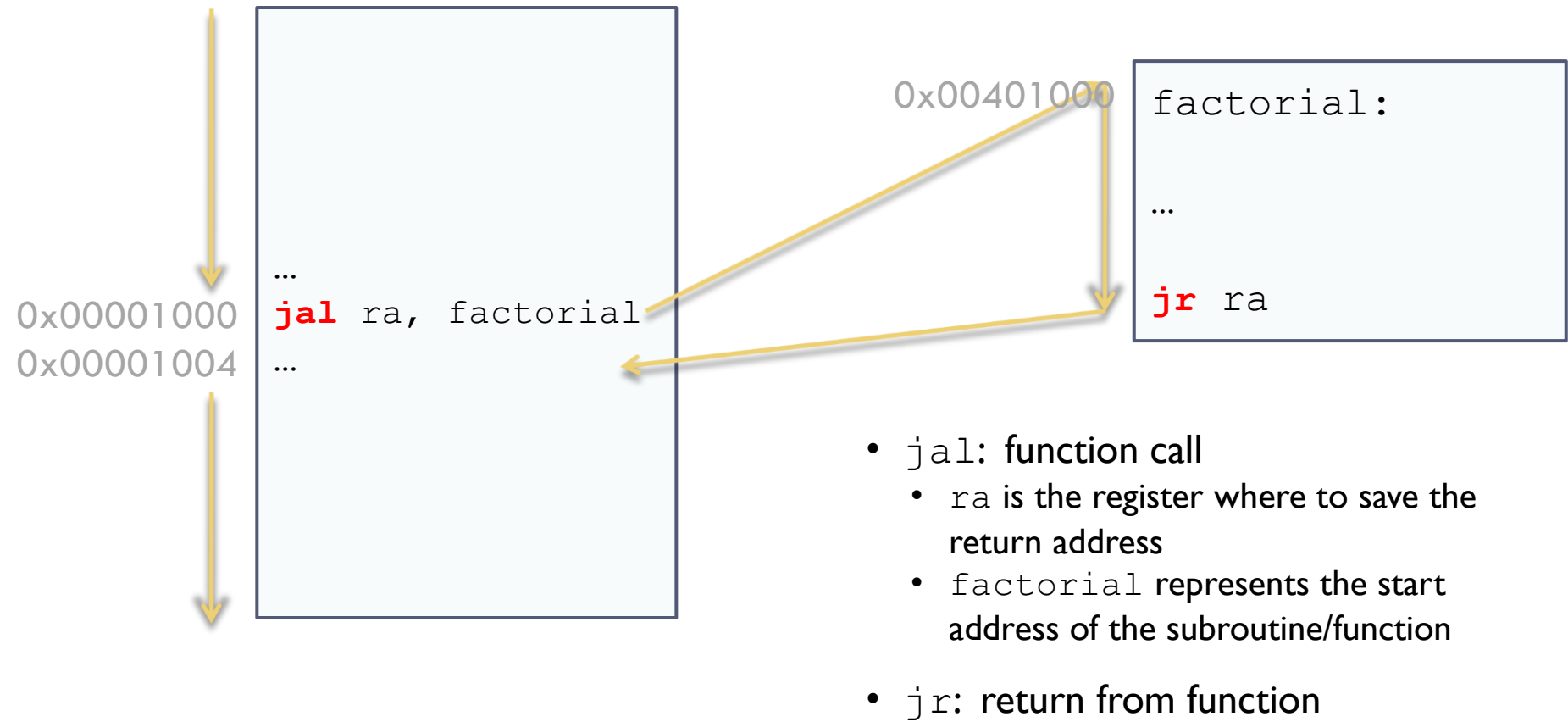
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    for (i=1;i<=x;i++) {  
        r*=i;  
    }  
    return r;  
}  
...  
r1 = factorial(3) ;  
...
```

- ▶ A high-level function (procedure, method, etc.) is a subprogram that:
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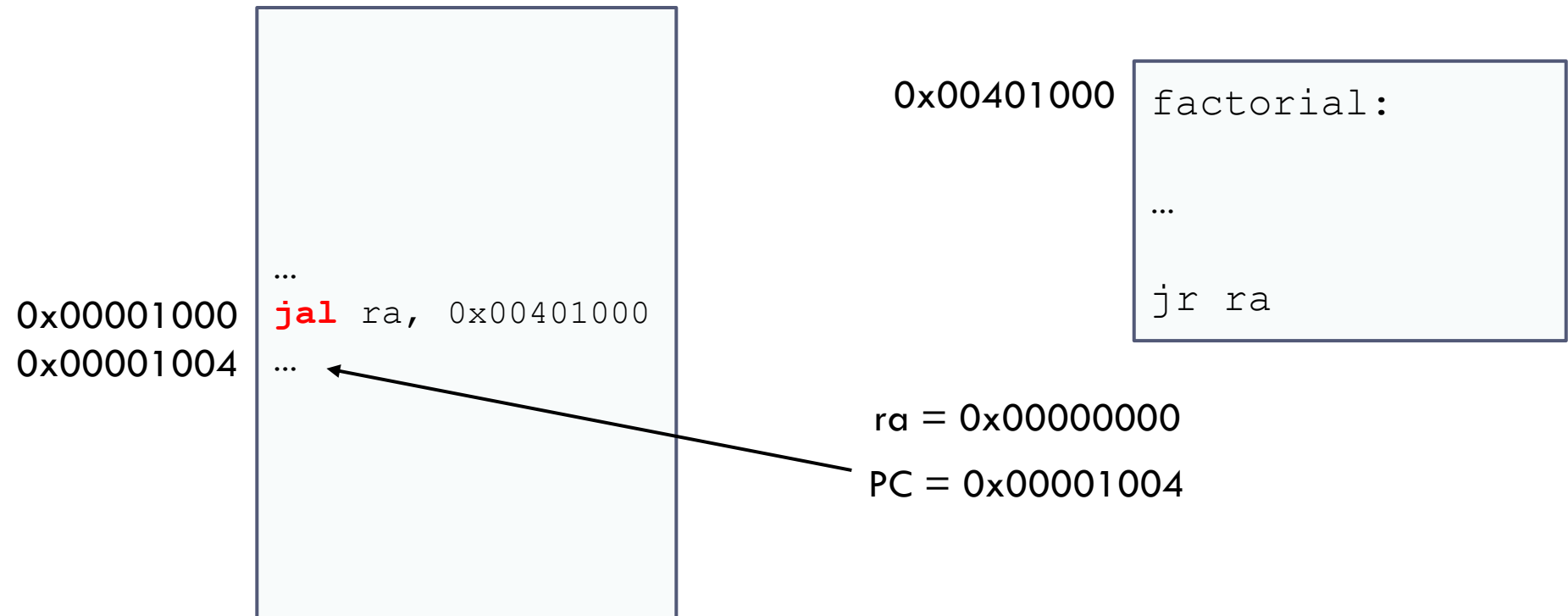
```
factorial:  
    mv    t0 a0  
    li    v0 1  
b1: beq   t0 zero f1  
    mul   v0 v0 t0  
    addi  t0 t0 -1  
    j     b1  
f1: jrr   ra  
...  
li    a0 3  
jal   ra factorial  
...
```

- ▶ In assembler, a function (subroutine) is associated with a label in the first instruction of the function
 - ▶ Symbolic name that denotes its starting address.
 - ▶ Memory address where the first instruction (of function) is located

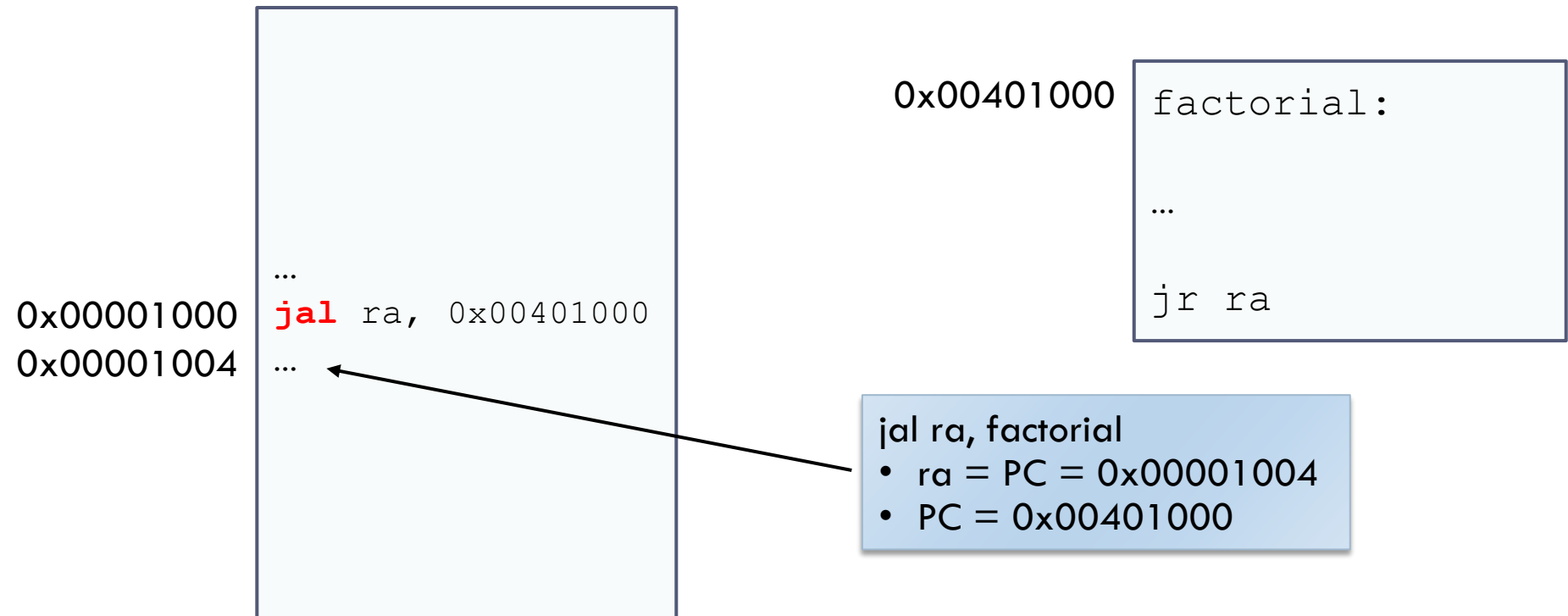
Function calls in RISC-V



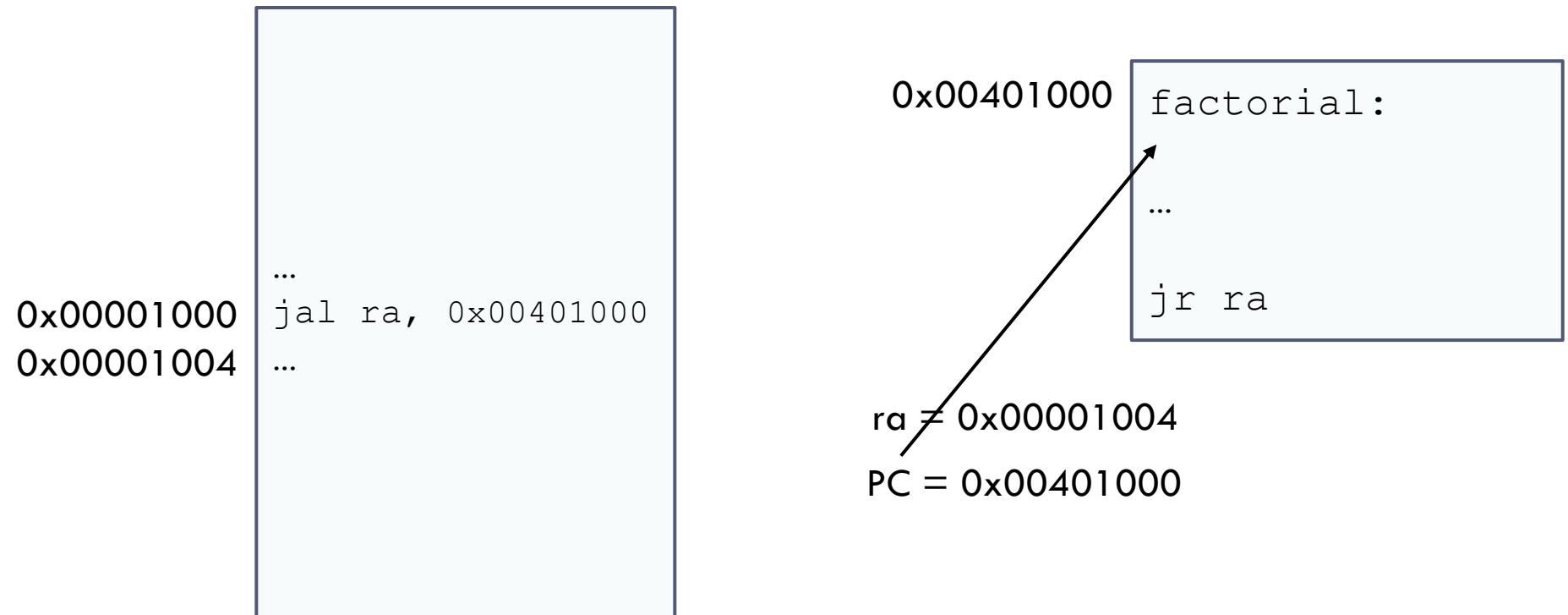
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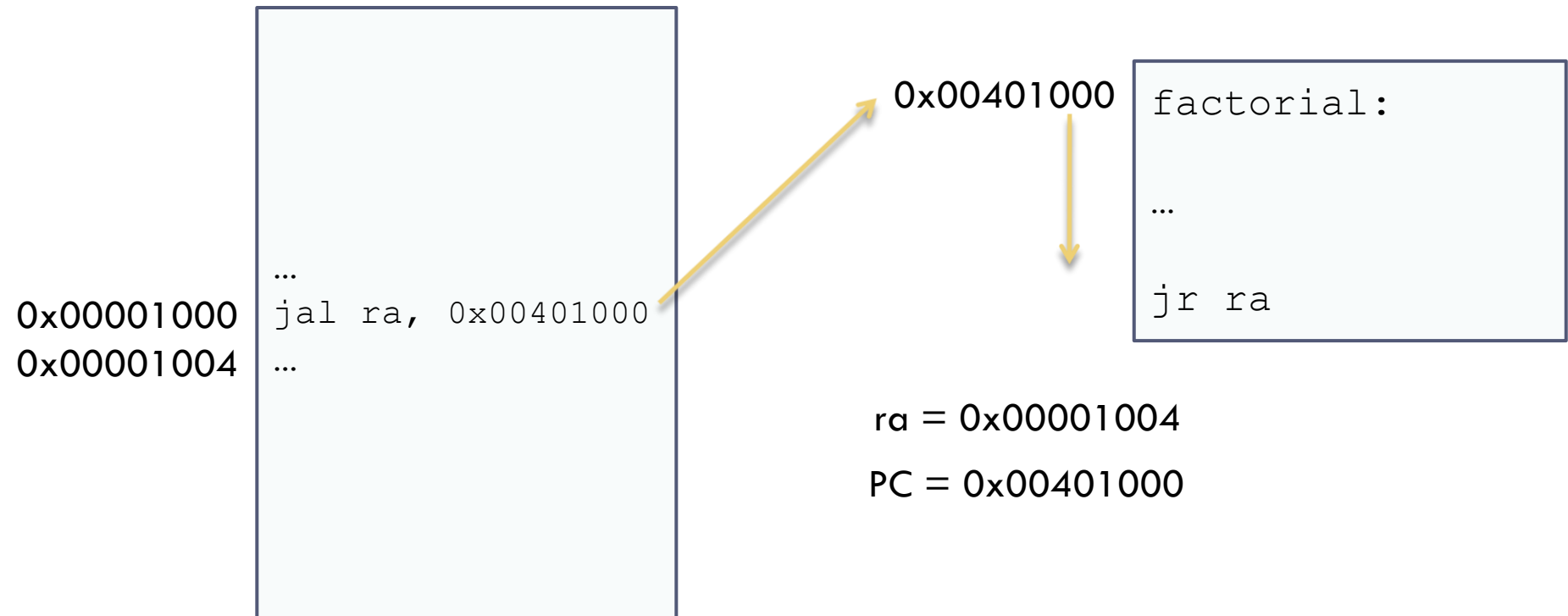
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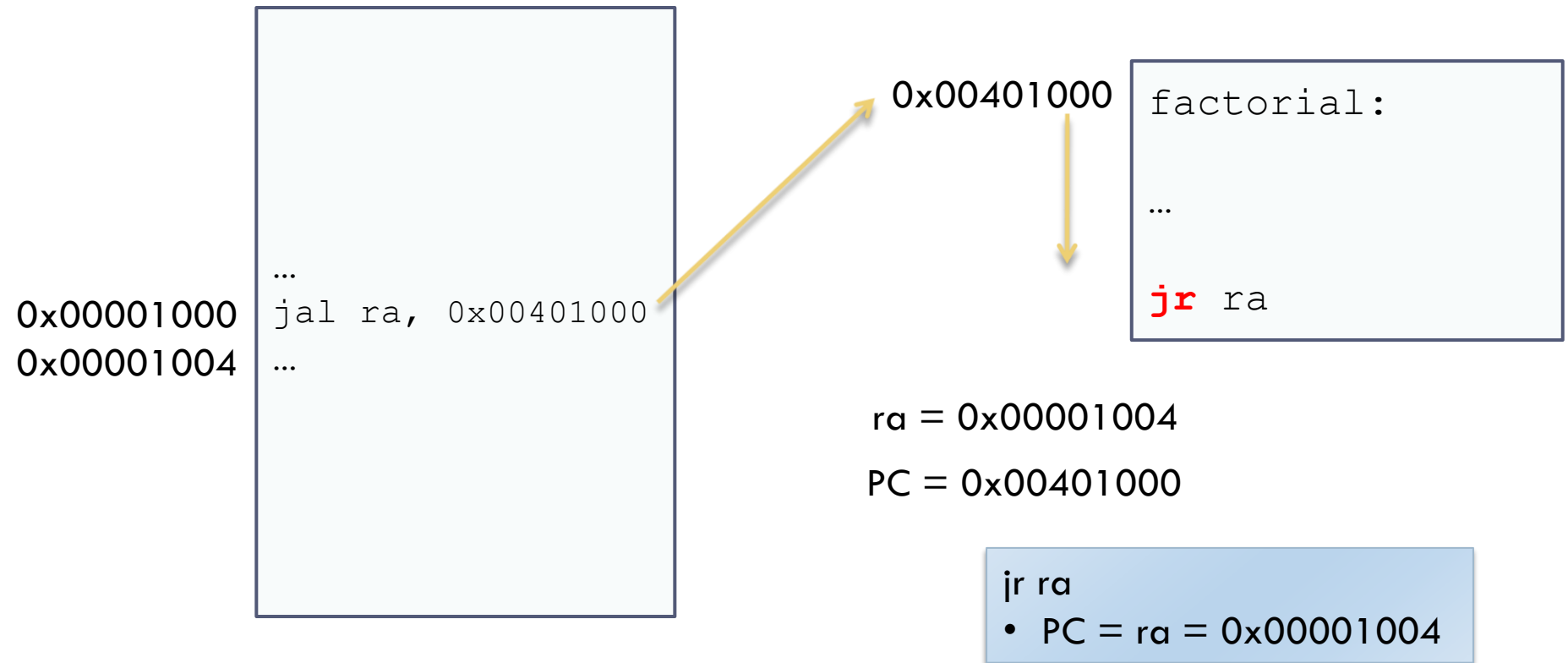
Function calls in RISC-V



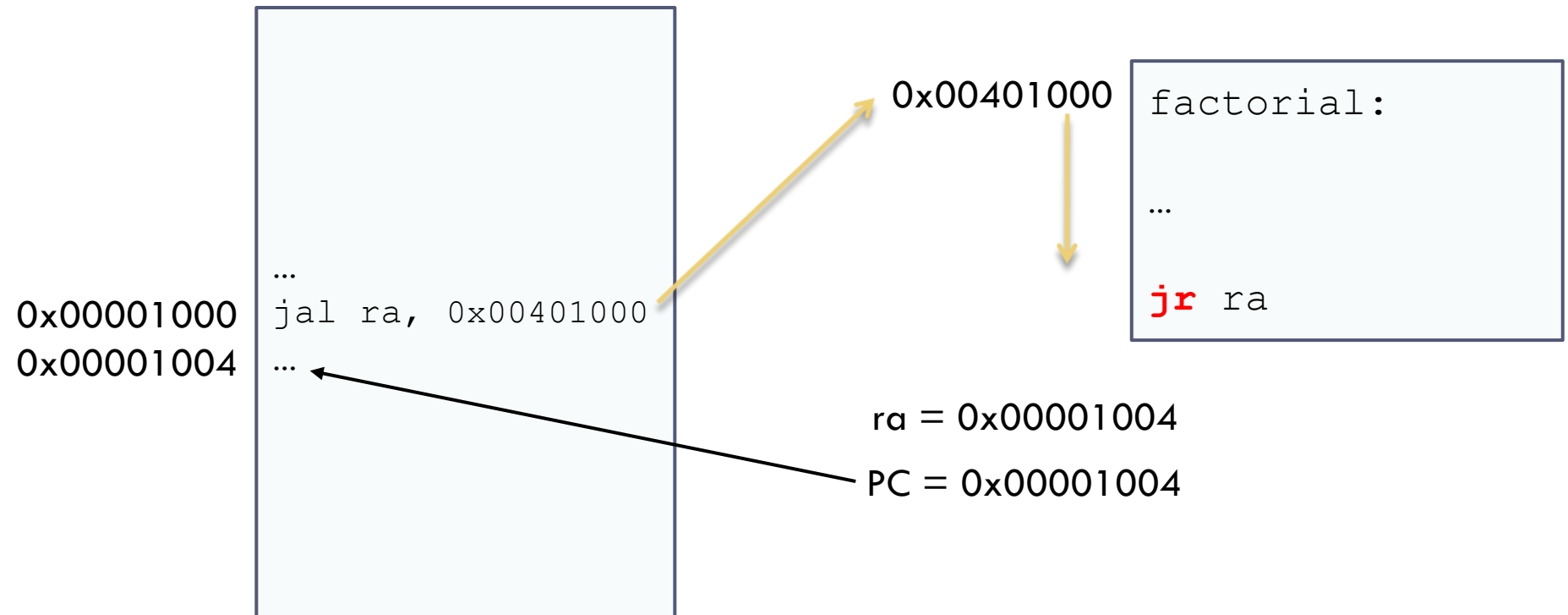
Function calls in RISC-V



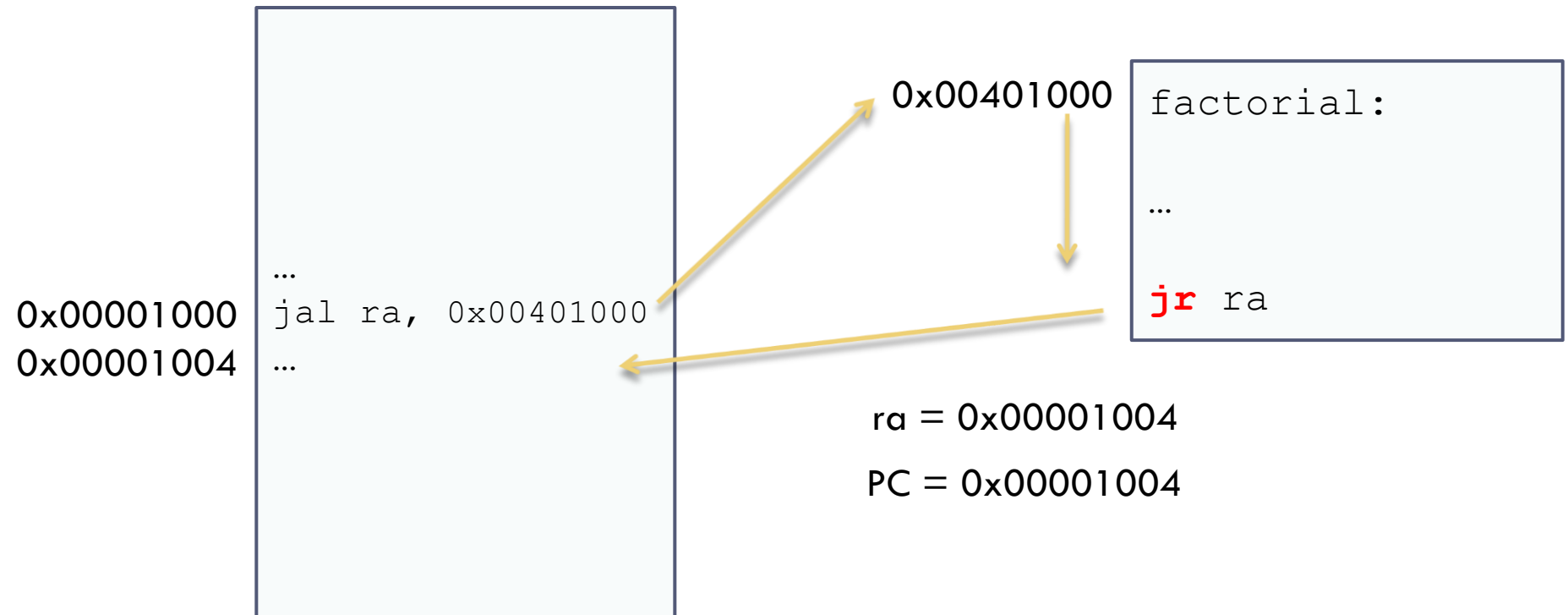
Function calls in RISC-V



Function calls in RISC-V



Function calls in RISC-V



jal/jr instructions

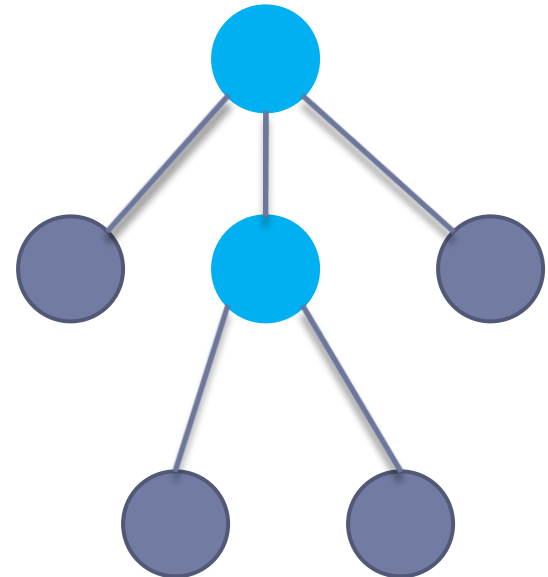
Subroutines / Functions		
jal reg2, label	reg2 = PC PC = label	• Loads the contents of PC into the reg2 register. When executing the jal instruction PC points to the first byte of the following instruction. • Calculates and loads into PC the memory address that the label represents. The next instruction to be executed will be the one pointed by PC.
jr reg1	PC = reg1	• Saves to PC the value stored in the reg1 registry.

Contents

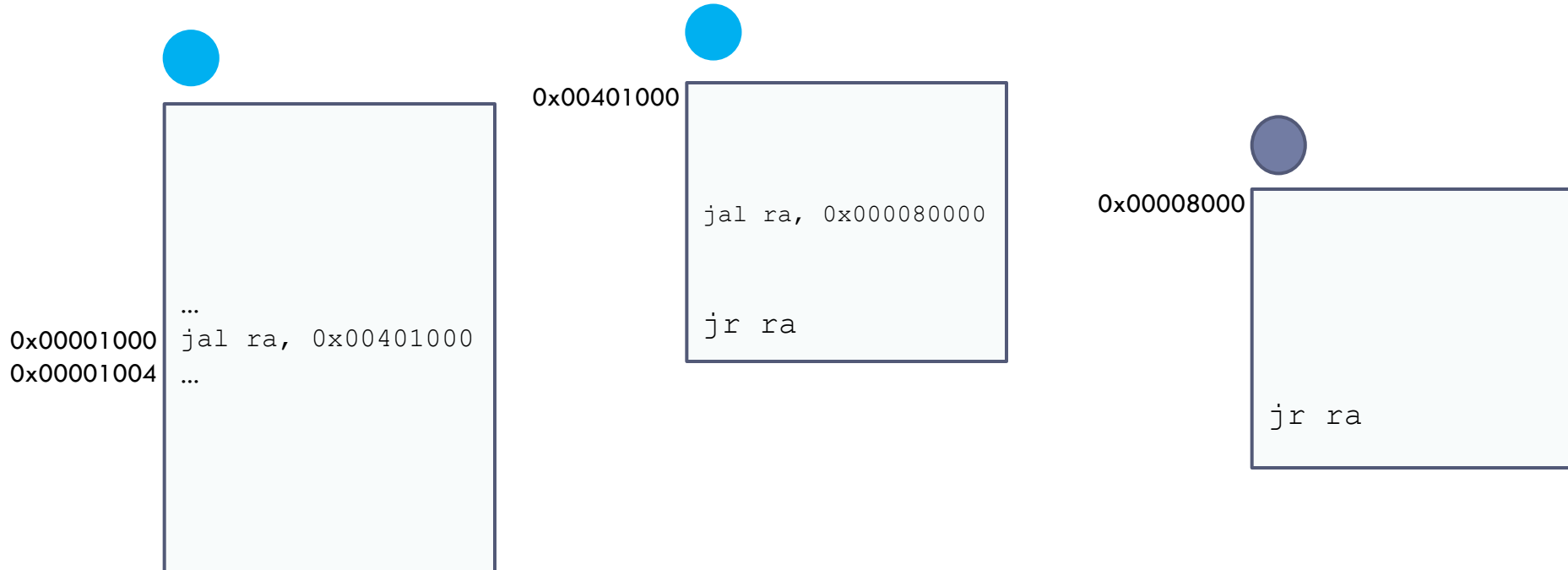
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Types of subroutines

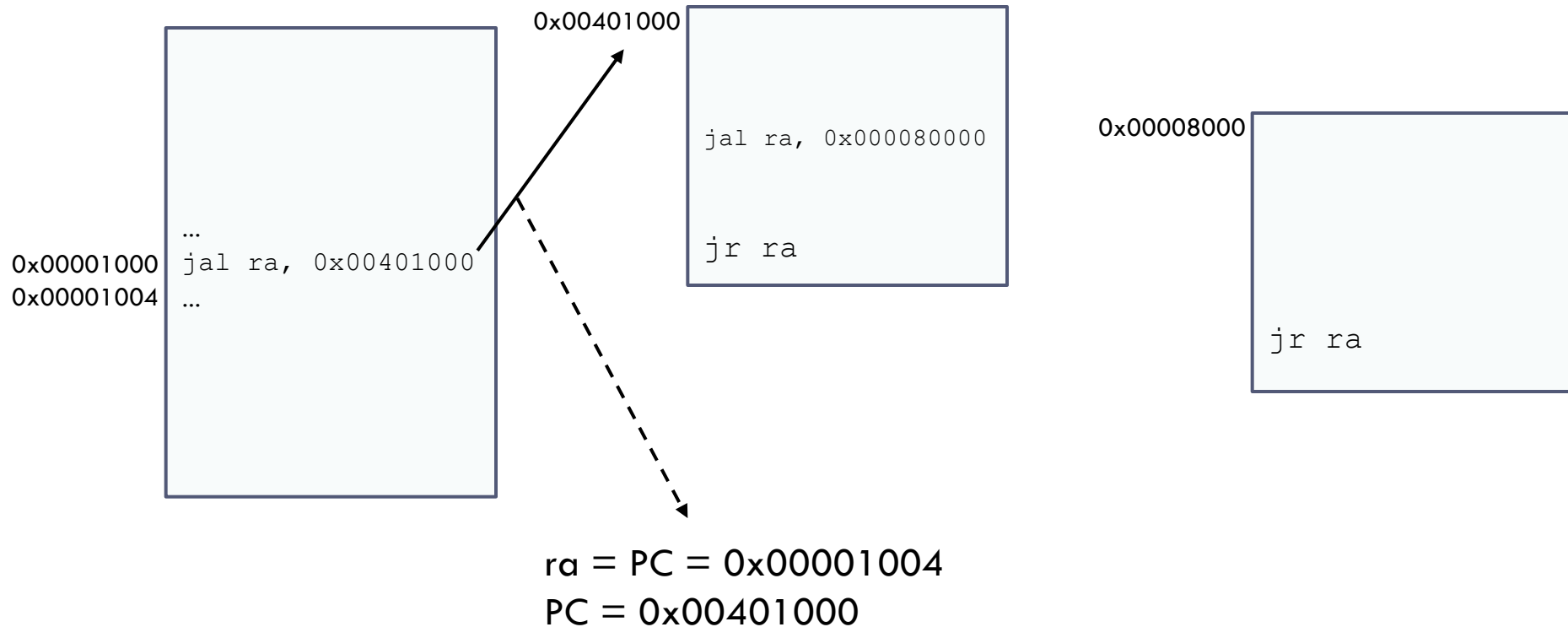
- **Terminal subroutine.**
 - ▶ **Does not** invoke any other subroutine.
- **Non-terminal subroutine.**
 - ▶ If you invoke any other subroutine.



Problem in non-terminal subroutines

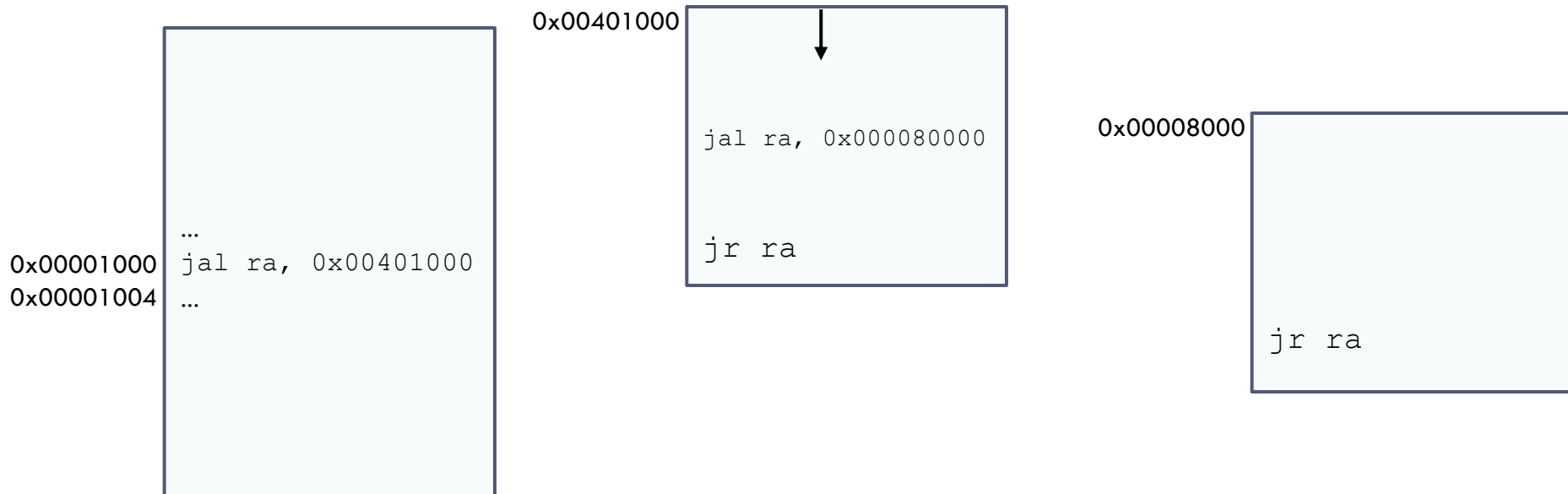


Problem in non-terminal subroutines



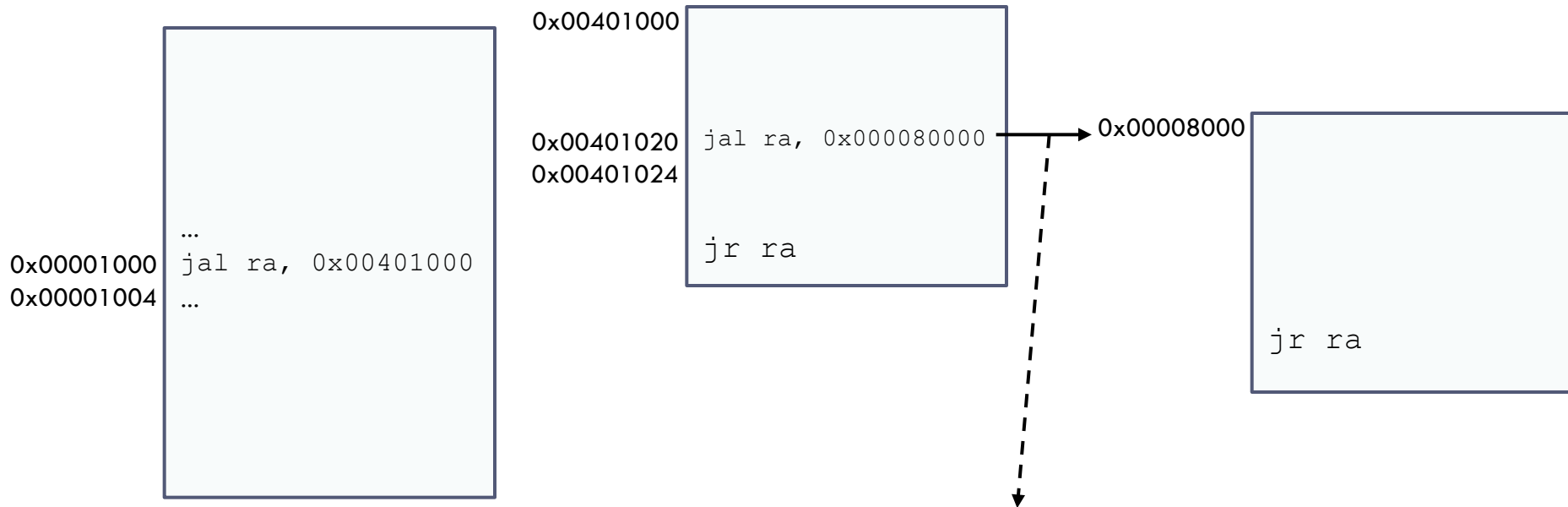
Return address ra = PC = 0x00001004

Problem in non-terminal subroutines



Return address `ra = PC = 0x00001004`

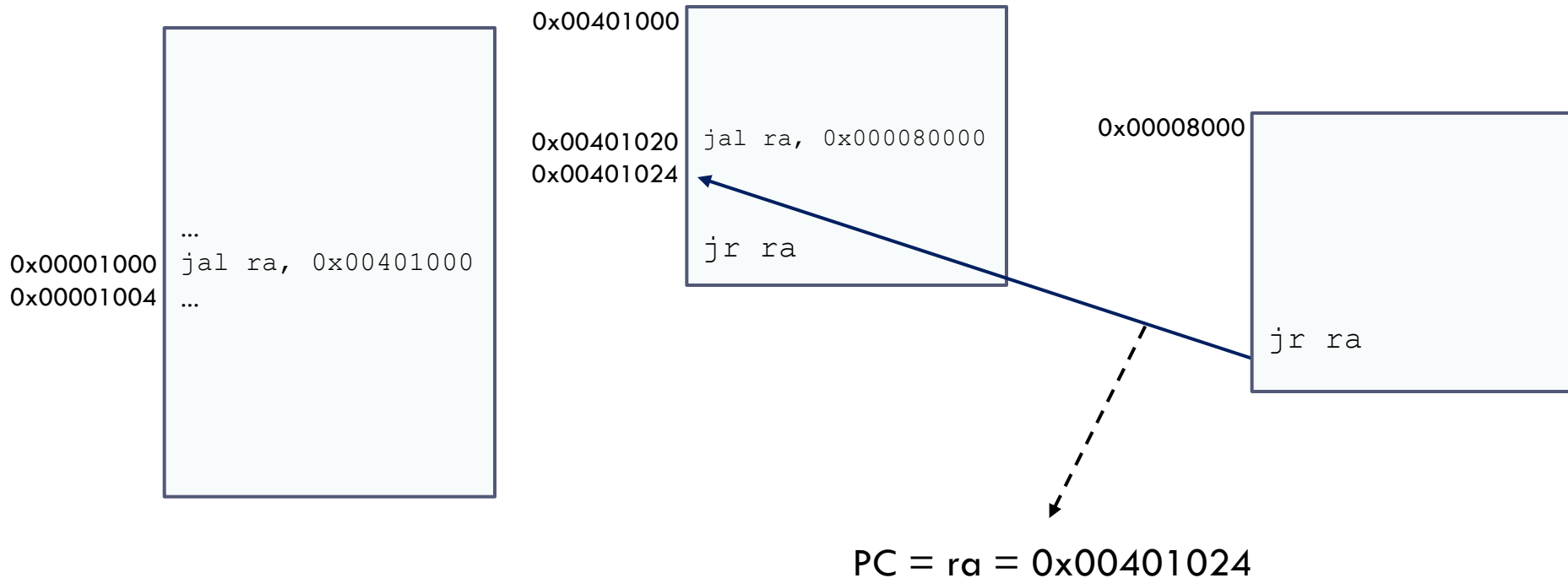
Problem in non-terminal subroutines



ra = PC = 0x00401024
PC = 0x00008000

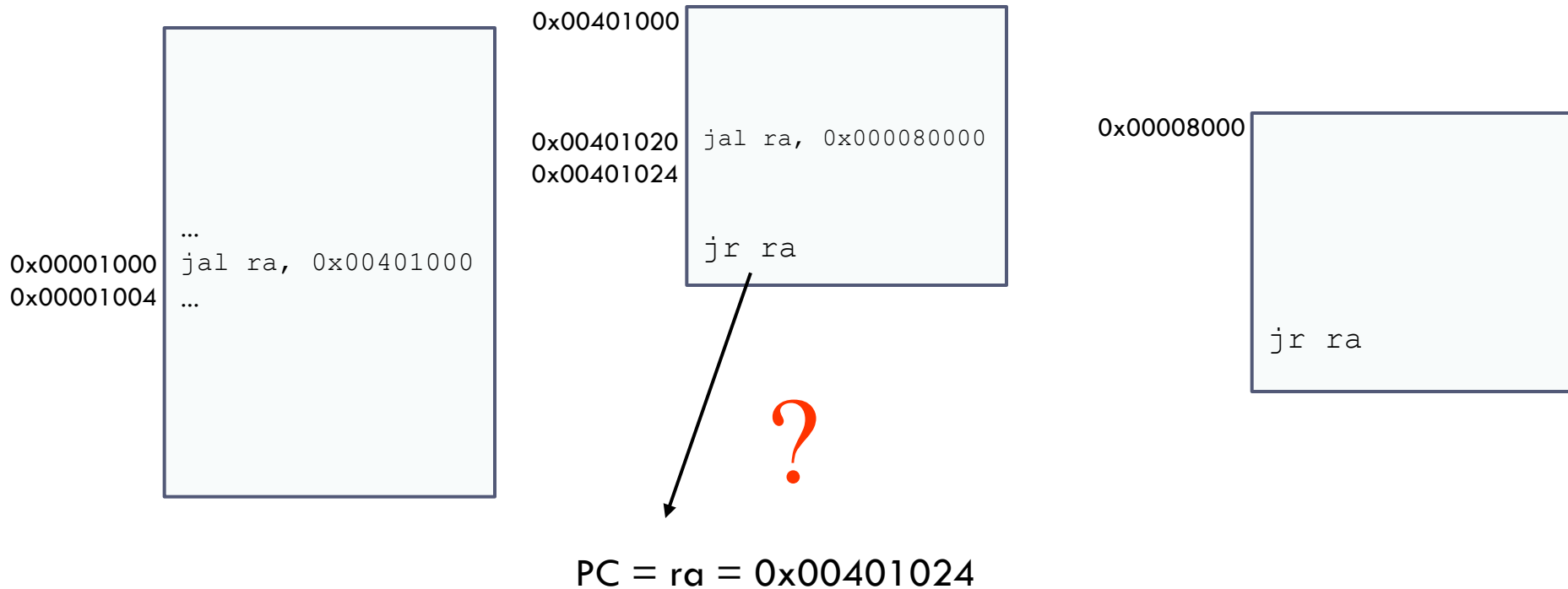
~~Return address ra = PC = 0x00001004~~

Problem in non-terminal subroutines



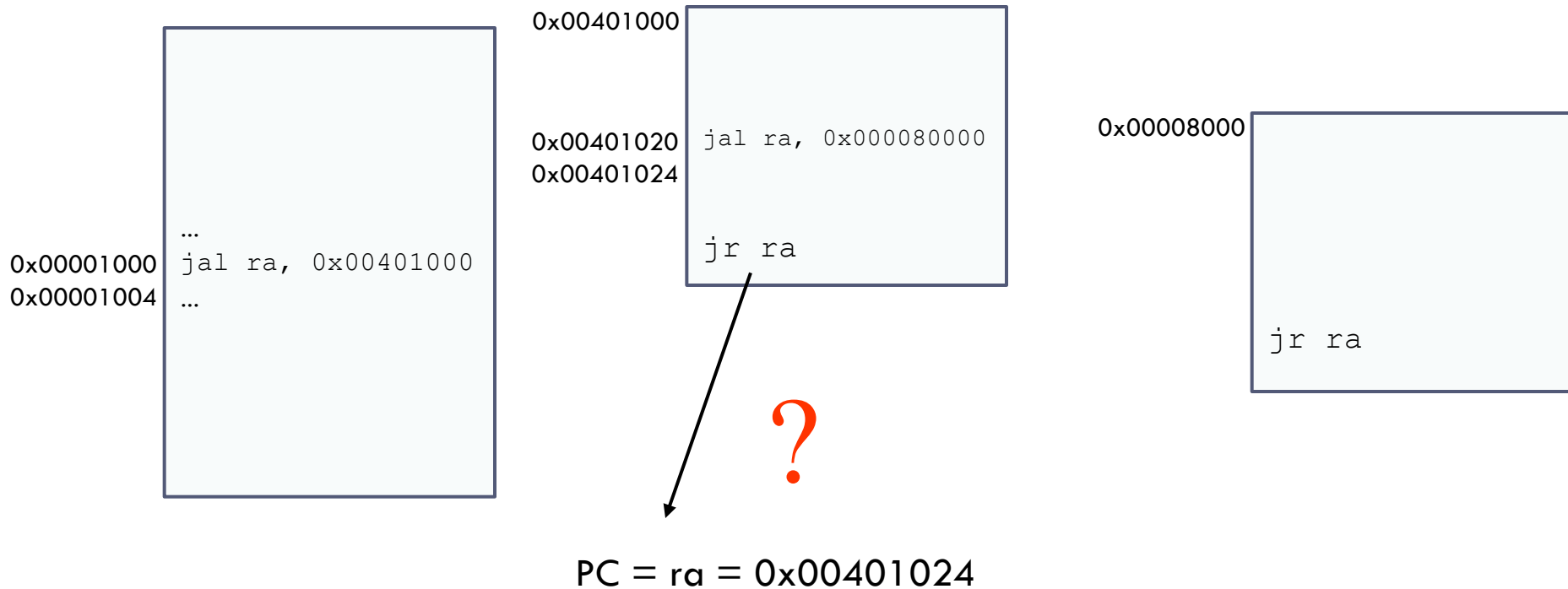
~~Return address ra = PC = 0x00001004~~

Problem in non-terminal subroutines



~~Return address ra = PC = 0x00001004~~

Problem in non-terminal subroutines



Return address has been lost !

Where to store the return address?

- ▶ Computers have two storage elements:
 - ▶ Registers
 - ▶ Memory
- ▶ Registers:
 - ▶ The number of registers is limited, so registers cannot be used (e.g.: recursive calls)
- ▶ Memory:
 - ▶ Return addresses are stored in main memory
 - ▶ In a program area called the **stack**

Program execution

Reminder

```
no_terminal:
```

```
    addi sp, sp, -4  
    sw   ra, 0(sp)
```

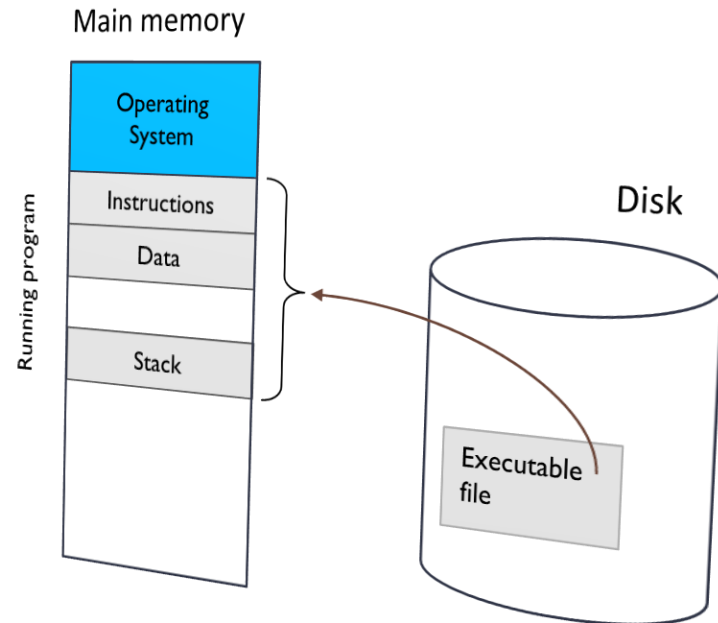
```
    li   t0, 8  
    li   s0, 9
```

```
    ...
```

```
    jal  ra, función
```

```
    ...
```

```
    lw   ra, 0(sp)  
    addi sp, sp, 4  
    jr   ra
```



Program execution

Reminder

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no_terminal:
```

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addi sp, sp, -4  
sw    ra, 0(sp)
```

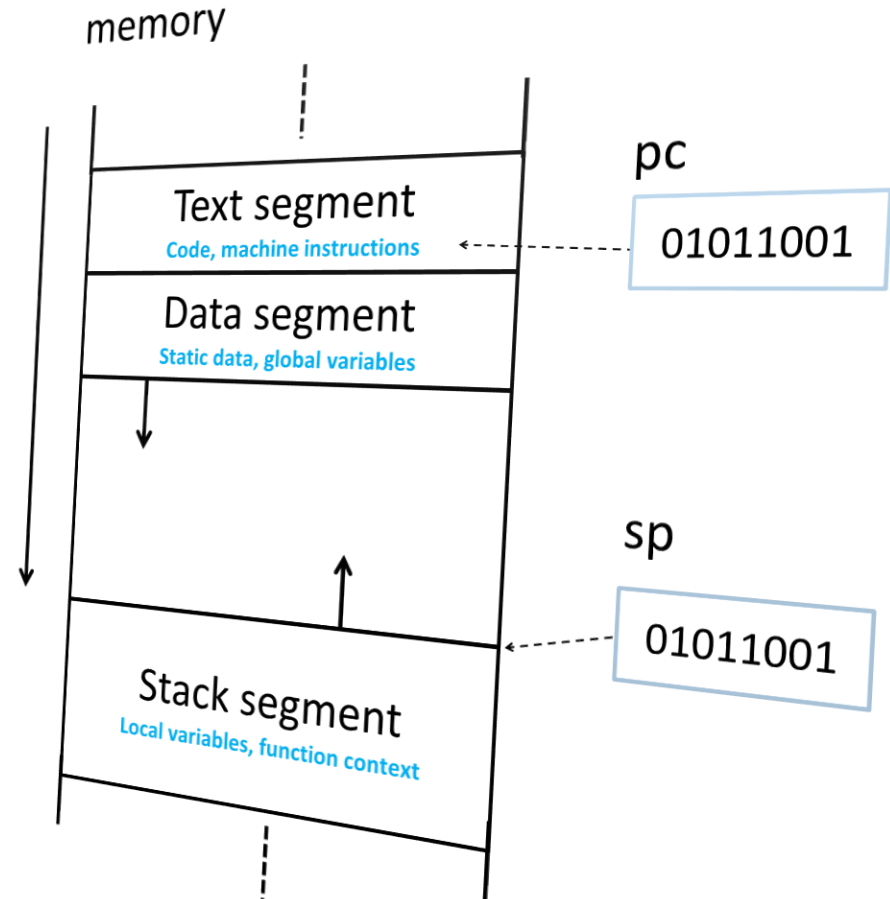
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li    t0, 8  
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```
...
```

```
jal   ra, función
```

```
...
```

```
lw    ra, 0(sp)  
addi sp, sp, 4  
jr    ra
```



Stack, jal y jr...

IMPORTANT!

`no_terminal:`

```
addi sp, sp, -4  
sw    ra, 0(sp)
```

← ra is saved in the stack at the beginning

```
li    t0, 8  
li    s0, 9
```

...

```
jal   ra, función
```

...

```
lw    ra, 0(sp)  
addi sp, sp, 4  
jr    ra
```

← the value is restored before "jr ra"

Program execution

Reminder

`no_terminal:`

```
addi sp, sp, -4  
sw    ra, 0(sp)
```

```
li    t0, 8  
li    s0, 9
```

...

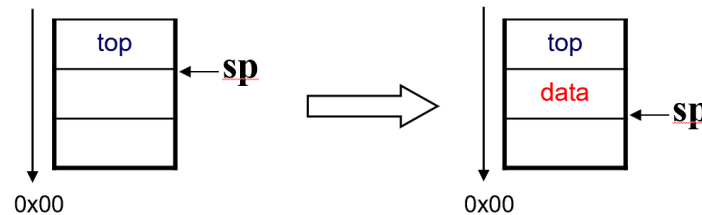
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```

...

```
lw    ra, 0(sp)  
addi sp, sp, 4  
jr    ra
```

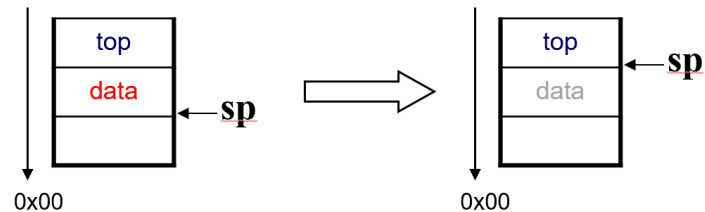
PUSH Reg

Stacks the contents of the record (data)



POP Reg

Unstack registry contents (data)
Copies data to the Reg registry



PUSH operation in RISC-V

Reminder

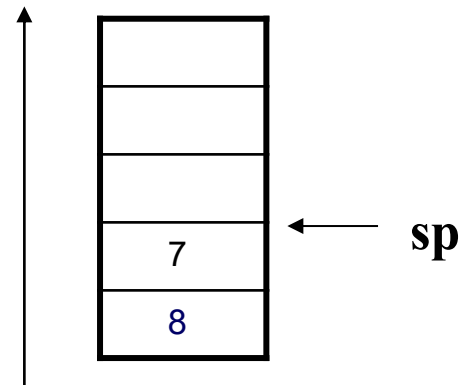
...

```
li    t2, 9
```

```
addi  sp, sp, -4
```

```
sw    t2, 0(sp)
```

...



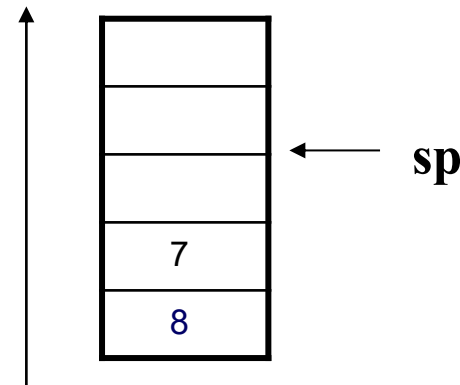
► Initial state:

- The stack pointer register (sp) points to the last element at the top of the stack
- The t2 register holds the value of 9

PUSH operation in RISC-V

Reminder

```
...  
li    t2, 9  
addi  sp, sp, -4  
sw    t2, 0(sp)  
...
```

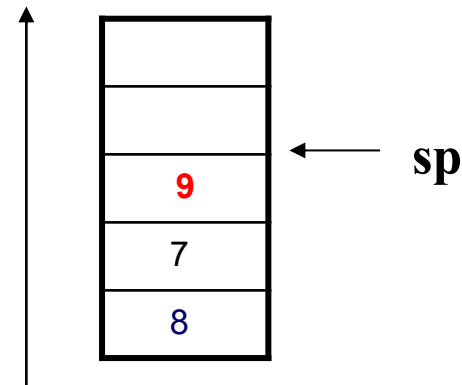


- ▶ Subtract 4 to stack pointer to insert a new word in the stack
 - ▶ `addi sp, sp, -4`

PUSH operation in RISC-V

Reminder

```
...  
li    t2, 9  
addi  sp, sp, -4  
sw    t2, 0(sp)  
...
```



- ▶ The contents of register t2 are inserted at the top of the stack:
 - ▶ `sw t2, 0(sp)`

POP operation in RISC-V₃₂

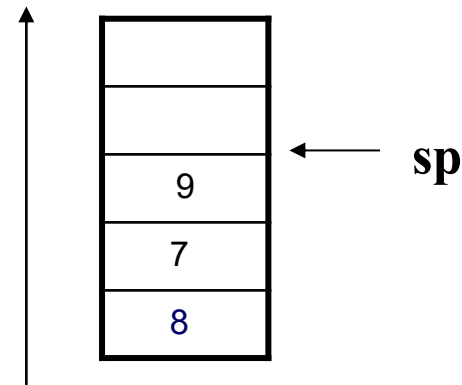
Reminder

...

```
lw    t2, 0(sp)
```

```
addi  sp, sp, 4
```

...



- ▶ The data stored at the top of the stack (9) is copied to t2.
 - ▶ lw t2, 0(sp)

POP operation in RISC-V

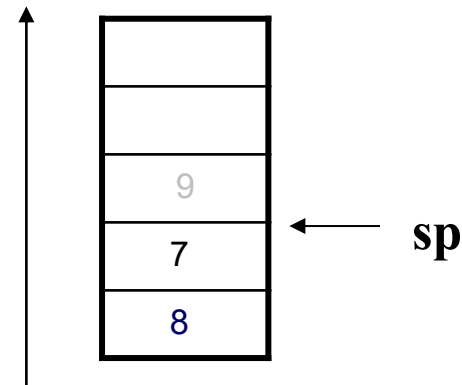
Reminder

...

```
lw    t2, 0(sp)
```

```
addi sp, sp, 4
```

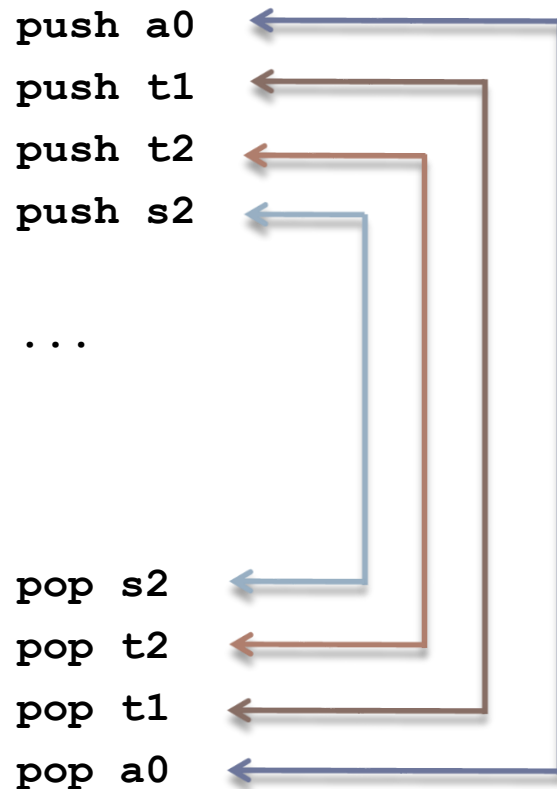
...



- ▶ The `sp` register is updated to point to the new top of the stack.
 - ▶ `addi sp, sp, 4`
- ▶ The unstacked data (9) is still in memory but will be overwritten in future PUSH (or similar memory access) operation

Stack: use of consecutive push and pop

a) unstacking in reverse order of stacking



Stack: use of consecutive push and pop

b) it is possible to add sums by operation or to join sums together

```
push a0
push t1
push t2
push s2
```

...

```
pop s2
pop t2
pop t1
pop a0
```

```
addi sp, sp, -4
sw a0, 0(sp)
addi sp, sp, -4
sw t1, 0(sp)
addi sp, sp, -4
sw t2, 0(sp)
addi sp, sp, -4
sw s2, 0(sp)
```

...

```
lw s2, 0(sp)
addi sp, sp, 4
lw t2, 0(sp)
addi sp, sp, 4
lw t1, 0(sp)
addi sp, sp, 4
lw a0, 0(sp)
addi sp, sp, 4
```

Stack: use of consecutive push and pop

b) it is possible to add sums by operation or to join sums together

```
push a0
push t1
push t2
push s2
```

...

```
pop s2
pop t2
pop t1
pop a0
```

```
addi sp, sp, -16
sw    a0, 12(sp)
sw    t1, 8(sp)
sw    t2, 4(sp)
sw    s2, 0(sp)
```

...

```
lw    s2, 0(sp)
lw    t2, 4(sp)
lw    t1, 8(sp)
lw    a0, 12(sp)
addi sp, sp, 16
```

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Parameters and registers agreement

```
no_terminal:
```

```
    addi sp, sp, -4  
    sw   ra, 0(sp)
```

```
    li   t0, 8  
    li   s0, 9
```

... ← In which registers are the parameters

```
    jal  ra, función
```

... ← passed and the results returned?

```
    lw   ra, 0(sp)  
    addi sp, sp, 4  
    jr   ra
```

Parameter passing agreement

- ▶ When programming in assembler, a convention is defined that specifies how arguments are passed and how registers are treated.
- ▶ Compilers define this convention for a given architecture.
- ▶ A simplified version of the conventions used by compilers will be used in this course.

Simplified agreement (RISC-V)

IMPORTANT!

▶ **Parameter passing**

- ▶ **Integer** parameters (char, int) are passed in **a0 ... a7**
 - ▶ If you need to pass more than eight parameters, first eight parameters in a0 ... a7 and the rest in the stack
- ▶ **Float** parameters are passed in **fa0 ... fa7**
 - ▶ If you need to pass more than eight parameters, the rest in the stack
- ▶ **Double** parameters are passed in **fa0 ... fa7**
 - ▶ If you need to pass more than eight parameters, the rest in the stack

▶ **Return on results**

- ▶ **a0** and **a1** are used for integer type values.
- ▶ **fa0** and **fa1** are used for float and double values.
- ▶ In case of structures or complex values, they must be left on the stack. The space is reserved by the calling function (caller).

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Parameters and registers agreement

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```

```
    li   t0, 8  
    li   s0, 9
```

```
    ...
```

```
    jal  ra, función
```

```
    ...
```

```
    lw   ra, 0(sp)  
    addi sp, sp, 4  
    jr  ra
```

What are the values of the t0
and s0 registers on return?



Convention for using registers (RISC-V)

IMPORTANT!

Name	Usage	Preserving value
zero	Constant 0	No
ra	Return address (subrutines)	Yes
sp	Stack pointer	Yes
gp	Global pointer	No
tp	Thread pointer	No
t0 ... t6	Temporal	No
s0/fp	Temporal / Frame pointer	Yes
s1 ... s11	Temporal	Yes
a0 ... a7	Argumento de entrada para rutinas	No

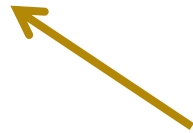
Name	Usage	Preserving value
ft0 ... ft11	Temporals	No
fs0 ... fs11	Temporals to save	Yes
fa0 ... fa1	Arguments/return	No
fa2 ... fa7	Arguments	No

Register agreement

```
li    t0, 8
li    s0, 9

li    a0, 7    # parameter
jal   ra, función
```

...



According to the agreement:

- **s0** will still be 9,
 - there is no guarantee that **t0** is 8
 - nor that **a0** is 7 after the execution of function.
- If we want **t0** to continue to be 8,
it must be saved on the stack
before each function call

Register agreement

```
li    t0, 8
li    s0, 9
```

```
addi  sp, sp, -4
sw    t0, 0(sp)
```

← It is saved in the stack before the call

```
li    a0, 7    # parameter
jal   ra, función
```

```
lw    t0, 0(sp)
addi  sp, sp, 4
```

← Value is restored after calling

```
...
```

Parameters and registers agreement

summary

no_terminal:

```
li  s0, 9
li  t0, 8
```

```
li  a0, 7    # parameter
jal ra, función
```

```
jr ra
```

Parameters and registers agreement

summary

no_terminal:

ra	DO preserve
sp	
s0 ... s11	

```
        addi sp, sp, -8
        sw   ra, 0(sp)
        sw   s0, 4(sp)
        li   s0, 9
        li   t0, 8

        li   a0, 7    # parameter
        jal  ra, función

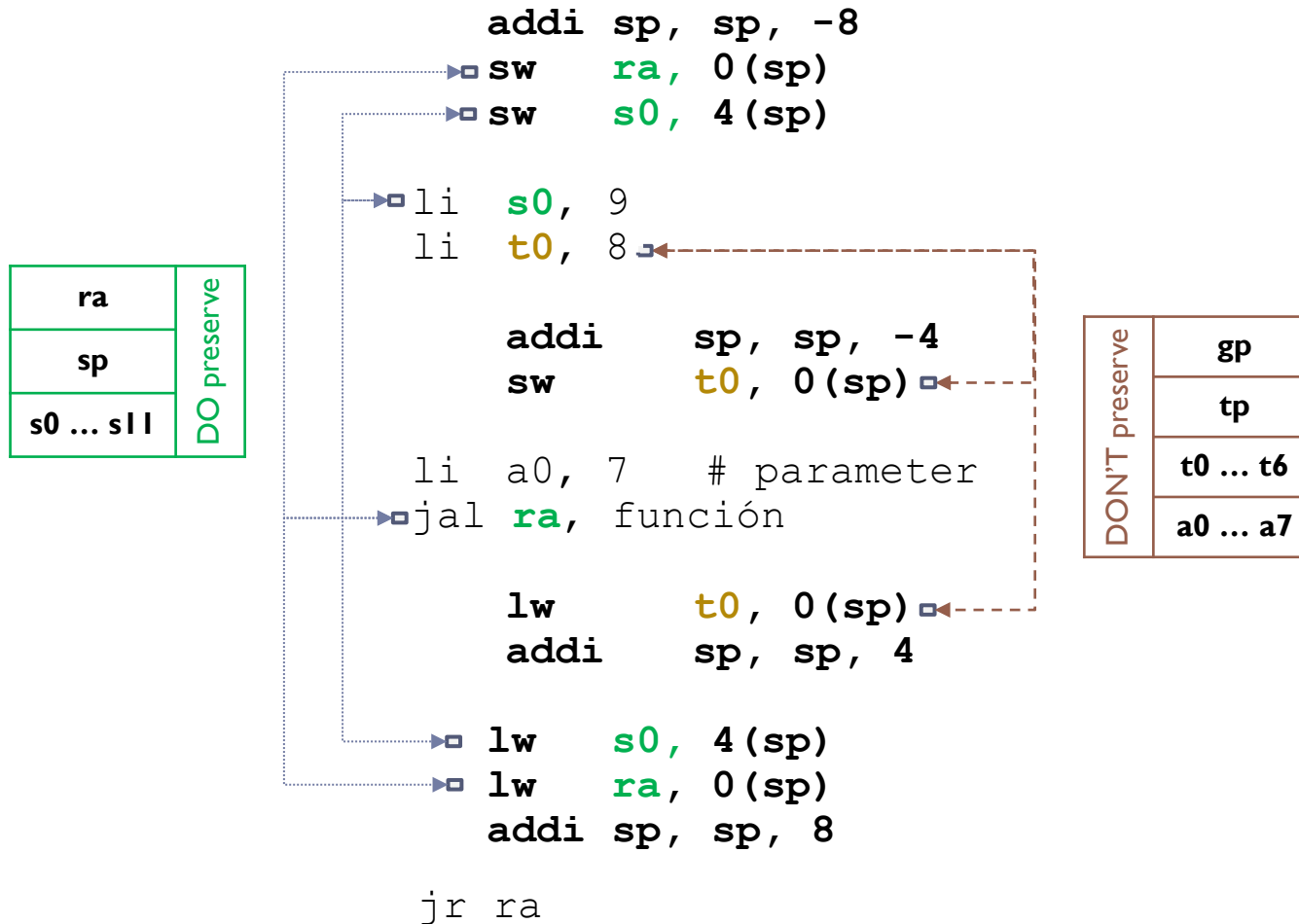
        lw   s0, 4(sp)
        lw   ra, 0(sp)
        addi sp, sp, 8

        jr   ra
```

Parameters and registers agreement

summary

no_terminal:



Parameters and registers agreement

summary

no_terminal:

- Local variables in stack.
- Example reserving 4 bytes (an integer)

ra	DO preserve
sp	
s0 ... s11	

```

    addi sp, sp, -12
    sw    ra, 0(sp)
    sw    s0, 4(sp)

```

```

    li    s0, 9
    li    t0, 8

```

```

    addi   sp, sp, -4
    sw     t0, 0(sp)

```

```

    li    a0, 7    # parameter
    jal   ra, función

```

```

    lw     t0, 0(sp)
    addi   sp, sp, 4

```

```

    lw     s0, 4(sp)
    lw     ra, 0(sp)
    addi   sp, sp, 12

```

```

    jr     ra

```

DON'T preserve	gp
	tp
	t0 ... t6
	a0 ... a7

Example

(1) Suppose a high-level language code

```
int main() {  
    int z;  
    z=factorial(5);  
    print_int(z);  
    .  
    .  
    .  
}  
  
int factorial(int x) {  
    int i;  
    int r=1;  
    for (i=1;i<=x;i++) {  
        r*=i;  
    }  
    return r;  
}
```



Example

(2) Analyze how to pass the arguments

- ▶ **arguments/parameters** are placed in a0 ... a7
 - ▶ `z=factorial(5)` has an input parameter in a0
- ▶ **Results** are collected in a0, a1
 - ▶ `z=factorial(5)` returns a result in a0
- ▶ If you need to pass more than eight parameters,
 - (1) the first eight in registers a0...a7 and
 - (2) rest on the stack
 - ▶ No more than eight parameters are required

Example

(3) Translate to assembly language

```
int main() {  
    int z;  
    z=factorial(5);  
    print_int(z);  
    . . .  
}  
  
int factorial(int x) {  
    int i;  
    int r=1;  
    for (i=1;i<=x;i++) {  
        r*=i;  
    }  
    return r;  
}
```

→ main: # factorial(5)
li a0, 5 # arg.
jal ra, factorial # invoke
mv a0, a0 # result
print_int(z)
li a7, 1
ecall
...

→ factorial: li s1, 1 #s1 for r
li s0, 1 #s0 for i
loop1: bgt s0, a0, end1
mul s1, s1, s0
addi s0, s0, 1
j loop1
end1: mv a0, s1 #result
jr ra

- The parameter is passed in a0
- The result is returned in a0

Example

(4) Analyze the registers modified (1/2)

```
int main() {  
    int z;  
    z=factorial(5);  
    print_int(z);  
}
```

main:

```
# factorial(5)  
li a0, 5 # arg.  
jal ra, factorial # invoke  
mv a0, a0 # result  
# print_int(z)  
li a7, 1  
ecall  
...
```

- main is non-terminal (there is a jal... that calls another subroutine).
- It is therefore modified ra

```
int factorial(int x) {  
    int i;  
    int r=1;  
    for (i=1; i<=x; i++) {  
        r*=i;  
    }  
    return r;  
}
```

factorial:

```
li s1, 1 #s1 for r  
li s0, 1 #s0 for i  
loop1: bgt s0, a0, end1  
mul s1, s1, s0  
addi s0, s0, 1  
j loop1  
end1: mv a0, s1 #result  
jr ra
```

Example

(4) Analyze the registers modified (2/2)

```
int main()
{
    int x;
    z=factorial(x);
    printf("%d", z);
    ...
}
```

- The factorial function uses (modifies) registers s0, s1, s2 and s3.
- If these registers are modified within the function, it could affect the function that made the call (the main function).
- Therefore, the factorial function must save the value of these registers on the stack at the beginning and restore them at the end.

```
int factorial(int x) {
    int i;
    int r=1;
    for (i=1; i<=x; i++) {
        r*=i;
    }
    return r;
}
```



```
...
factorial: li    s1, 1      #s1 for r
           li    s0, 1      #s0 for i
loop1:    bgt    s0, a0, end1
           mul    s1, s1, s0
           addi   s0, s0, 1
           j      loop1
end1:     mv     a0, s1      #result
           jr     ra
```

Example

(5) Store registers in stack (1/2)

```
int main() {
    int z;
    z=factorial(5);
}
```

- It is necessary to save ra.
- The main routine is non-terminal
- Neither s0...s11 nor t0...t6 should be saved

```
int factorial(int x) {
    int i;
    int r=1;
    for (i=1;i<=x;i++) {
        r*=i;
    }
    return r;
}
```

main:

```
addi sp, sp, -4
sw    ra, 0(sp)
# factorial(5)
li    a0, 5          # arg.
jal   ra, factorial  # invoke
mv    a0, a0         # result
# print_int(z)
li    a7, 1
ecall
```

...

```
lw    ra, 0(sp)
add   sp, sp, 4
jr    ra
```

factorial:

```
addi   sp, sp, -8
sw     s0, 4(sp)
sw     s1, 0(sp)
li     s1, 1          # s1 para r
li     s0, 1          # s0 para i
loop1: bgt    s0, a0, end1
mul     s1, s1, s0
addi    s0, s0, 1
j       loop1
end1:   mv     a0, s1  # result
lw      s1, 0(sp)
lw      s0, 4(sp)
addi    sp, sp, 8
jr      ra
```

Example

(5) Store registers in stack (2/2)

```
int main() {
    int z;
    z=factorial(5);
    print_int(z);
    . . .
}
```

main:

```
addi sp, sp, -4
sw    ra, 0(sp)
# factorial(5)
li    a0, 5          # arg.
jal   ra, factorial  # invoke
mv    a0, a0         # result
# print_int(z)
li    a7, 1
ecall

...
lw    ra, 0(sp)
add   sp, sp, 4
jr    ra
```

- It is not necessary to save ra.
- The main routine is terminal
- s0, s1 stored on stack (they are modified)
- Not necessary if using t0 and t1

```
    r*=1;
}
return r;
}
```

```
factorial:
addi   sp, sp, -8
sw     s0, 4(sp)
sw     s1, 0(sp)
li     s1, 1      # s1 para r
li     s0, 1      # s0 para i
loop1: bgt     s0, a0, endl
mul     s1, s1, s0
addi    s0, s0, 1
j       loop1
endl:   mv     a0, s1  # result
lw      s1, 0(sp)
lw      s0, 4(sp)
addi    sp, sp, 8
jr      ra
```

Example 2

```
int main()
{
    int z;

    z=f1(5, 2);

    pint(z);
}

int f1(int a, int b)
{
    int r;

    r = a+a+f2(b);
    return r;
}

int f2(int c)
{
    int s;

    s = c * c * c;
    return s;
}
```

The diagram illustrates the flow of function calls in the provided C code. A yellow arrow points from the boxed 'f1' in the line 'z=f1(5, 2);' within the 'main' function to the 'f1' function definition. Another yellow arrow points from the boxed 'f2' in the line 'r = a+a+f2(b);' within the 'f1' function to the 'f2' function definition. Small white boxes with black borders highlight the function names 'f1' and 'f2' at the call sites.

Example 2. Body of main (1/3)

```
int main()
{
    int z;

    z=f1(5, 2);

    pint(z);
}
```



```
main:
    li    a0, 5      # first argument
    li    a1, 2      # second argument
    jal   ra, f1      # call
                          # result (a0)

    li    a7, 1
    ecall                # system call
                          # to print int

    jr    ra
```

Example 2. Analysis of main (2/3)

```
int main()
{
    int z;

    z=f1(5, 2);

    pint(z);
}
```

main:

```
li    a0, 5      # first argument
li    a1, 2      # second argument
jal   ra, f1     # call
                        # result (a0)

li    a7, 1
ecall                # system call
                        # to print int
```

- The parameters are passed in a0 and a1.
- The result is returned in a0
- Non-terminal routine (calls another routine)

Example 2. Adjustment of main (3/3)

```
int main()
{
    int z;

    z=f1(5, 2);

    pint(z);
}
```

main:

```
addi sp, sp, -4
sw    ra, 0(sp)
```

```
li    a0, 5      # first argument
li    a1, 2      # second argument
jal   ra, f1     # call
                        # result (a0)
```

```
li    a7, 1
ecall                        # system call
                        # to print int
```

```
lw    ra, 0(sp)
addi sp, sp, 4
jr    ra
```

Example 2. Body of f1 (1/3)

```
int f1 (int a, int b)
{
    int r;

    r = a + a + f2(b);
    return r;
}
```



```
f1: add    s0, a0, a0

      mv    a0, a1
      jal   ra, f2
      add   a0, s0, a0

      jr    ra
```

```
int f2(int c)
{
    int s;

    s = c * c * c;
    return s;
}
```

Example 2. Analysis of f1 (2/3)

```
int f1 (int a, int b)
{
    int r;

    r = a + a + f2(b);
    return r;
}
```



```
f1: add    s0, a0, a0

      mv    a0, a1
      jal   ra, f2
      add   a0, s0, a0

      jr    ra
```

```
int f2(int c)
{
    int s;

    s = c * c * c;
    return s;
}
```

- f1 modifies s0 and ra, therefore they are saved on the stack.
- The register ra is modified in the instruction "jal ra f2".
- The register a0 is modified by passing the argument to f2, but by convention the function f1 does not have to save it on the stack only if it uses it after making the call to f2

Example 2. Body of f1 storing in the stack the registers that are modified (3/3)

```
int f1 (int a, int b)
{
    int r;

    r = a + a + f2(b);
    return r;
}
```



```
int f2(int c)
{
    int s;

    s = c * c * c;
    return s;
}
```

```
f1: addi    sp, sp, -8
     sw     s0, 4(sp)
     sw     ra, 0(sp)
```

```
     add    s0, a0, a0
     mv     a0, a1
     jal    ra, f2
     add    a0, s0, a0
```

```
     lw     ra, 0(sp)
     lw     s0, 4(sp)
     addu    sp, sp, 8
```

```
     jr     ra
```

Example 2. Body and analysis of f2

```
int f1 (int a, int b)
{
    int r;

    r = a + a + f2(b);
    return r;
}
```

```
int f2(int c)
{
    int s;

    s = c * c * c;
    return s;
}
```



```
f2: mul t0, a0, a0
     mul a0, t0, a0
     jr  ra
```

- The function f2 does not modify the register ra because it does not call any other function.
- The register t0 does not need to be stored because its value is not to be preserved according to convention.

Contents

- ▶ Basic concepts on assembly programming
- ▶ RISC-V 32 assembly language, memory model and data representation
- ▶ Instruction formats and addressing modes
- ▶ Procedure calls and stack convention
 - ▶ How do you call a function/subroutine?
 - ▶ Where is the return address stored in non-terminal routines?
 - ▶ What is the parameter passing convention?
 - ▶ What is the register use agreement?
 - ▶ What are the local variables like? (activation log)

Activation log

stack frame

- ▶ The **stack frame** or **activation register** is the mechanism used by the compiler to activate functions in high-level languages.
- ▶ The stack frame is built on the stack by the calling procedure/function and the called procedure/function.

Frame pointer

- ▶ The stack frame stores:
 - ▶ The **parameters entered** by the calling procedure, **if necessary**.
 - ▶ The **registers saved** by the function (including the `ra` register in case of non-terminal procedures/functions).
 - ▶ **Local variables**.

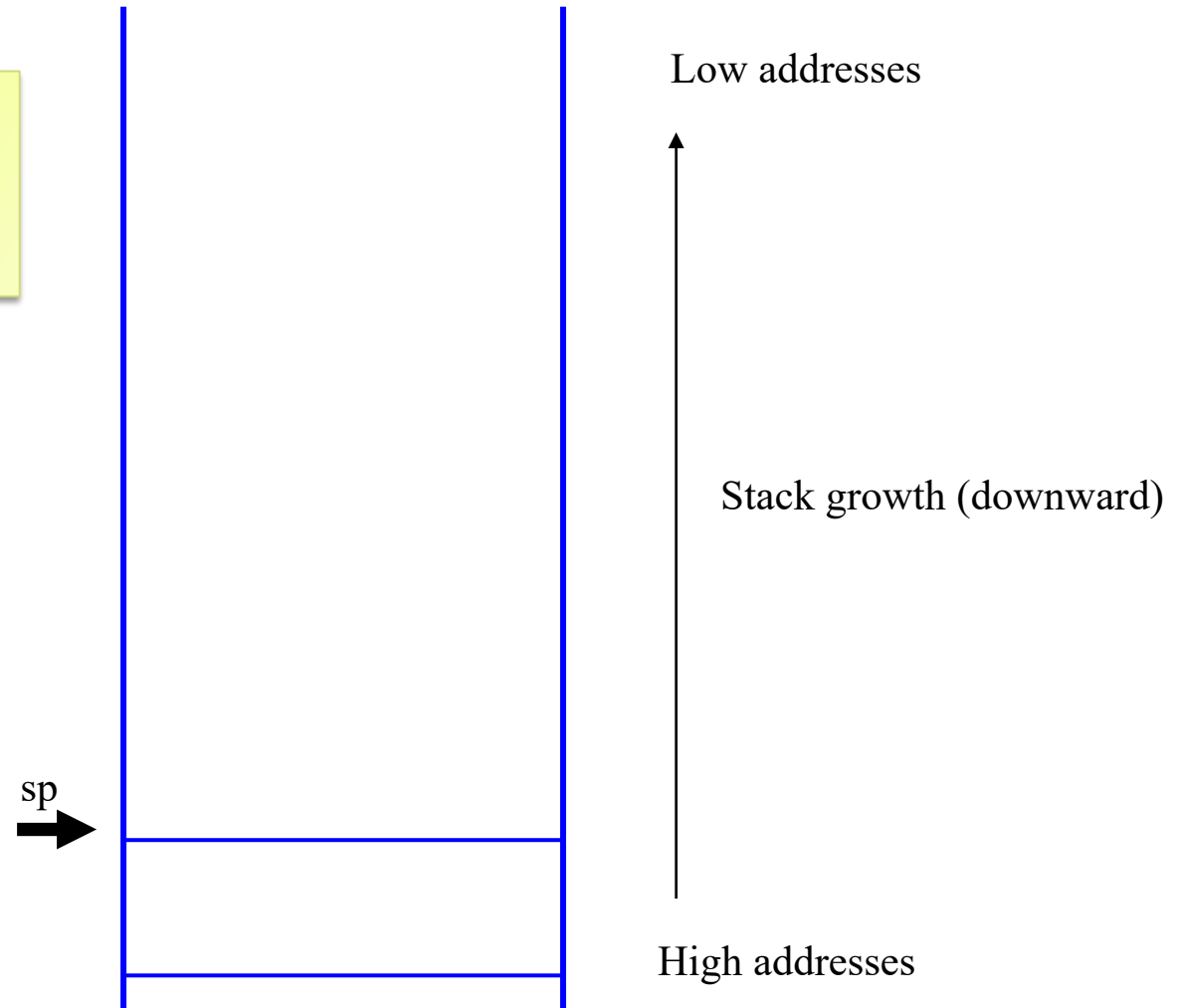
General function call steps

simplified version

Caller function	Calle function
Save the registers not preserved across the call ($t_x, a_x, ..$)	
Parameter passing + (if needed) allocation of space for values to be returned	
Make de call (jal)	
	Stacking frame allocation
	Save registers (ra, s_x)
	Function execution
	Restoring saved values
	Copy values to be returned in the space reserved by the caller
	Stack frame release (calle part)
	Return from function (jr ra)
Get returned values	
Restoration of saved registers, freeing the reserved stack space	

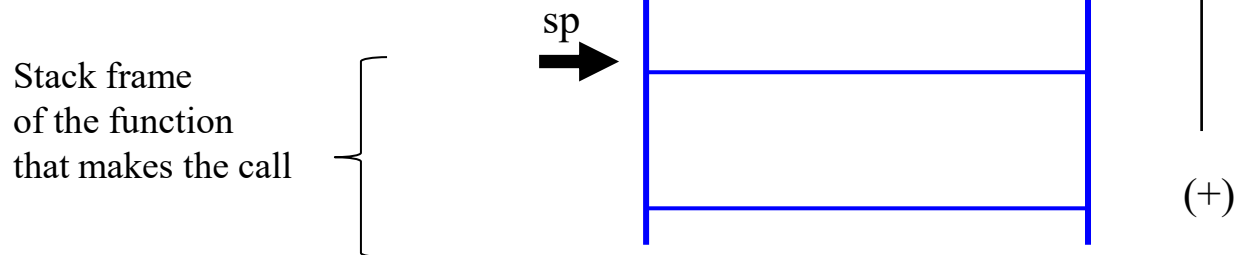
Construction of the stack frame caller function

The RISC-V convention
will not be strictly
followed for the sake of
simplicity



Construction of the stack frame caller function

Initial situation before
calling a function



Construction of the stack frame caller function

Saving registers

A function can modify
any register a_x y t_x

Stack frame
of the function
that makes the call

sp
→

Saved registers

Example:

```
li    t0, 4
li    t1, 8
li    a0, 5
jal   ra, funcion
```

```
mv    s2, t0
```

What is the value
of t0 and t1?

Construction of the stack frame caller function

Saving registers

A function can modify
any register a_x y t_x

To preserve their value,
the calling subroutine must save
the values of these registers
on the stack

Stack frame
of the function
that makes the call

sp
→

Saved registers

Example:

```
li    t0, 4
li    t1, 8
li    a0, 5
jal   ra, funcion
```

```
mv    s2, t0
```

What is the value
of t0 and t1?

Construction of the stack frame

caller function

Saving registers

A function can modify any register a_x y t_x

To preserve their value, the calling subroutine must save the values of these registers on the stack

Stack frame of the function that makes the call

sp
→

Saved registers

Example:

```
sub    sp, sp, 8
sw     t0, 0(sp)
sw     t1, 4(sp)

li     a0, 5
jal    ra, funcion
```

Construction of the stack frame caller function

Saving registers

A function can modify
any register a_x y t_x

To preserve their value,
the calling subroutine must save
the values of these registers
on the stack

- they will have to be restored
later.

Stack frame
of the function
that makes the call

sp
→

Saved registers

Example:

```
sub    sp, sp, 8
sw     t0, 0(sp)
sw     t1, 4(sp)

li     a0, 5
jal    ra, funcion

lw     t0, 0(sp)
lw     t1, 4(sp)
add    sp, sp, 8
```

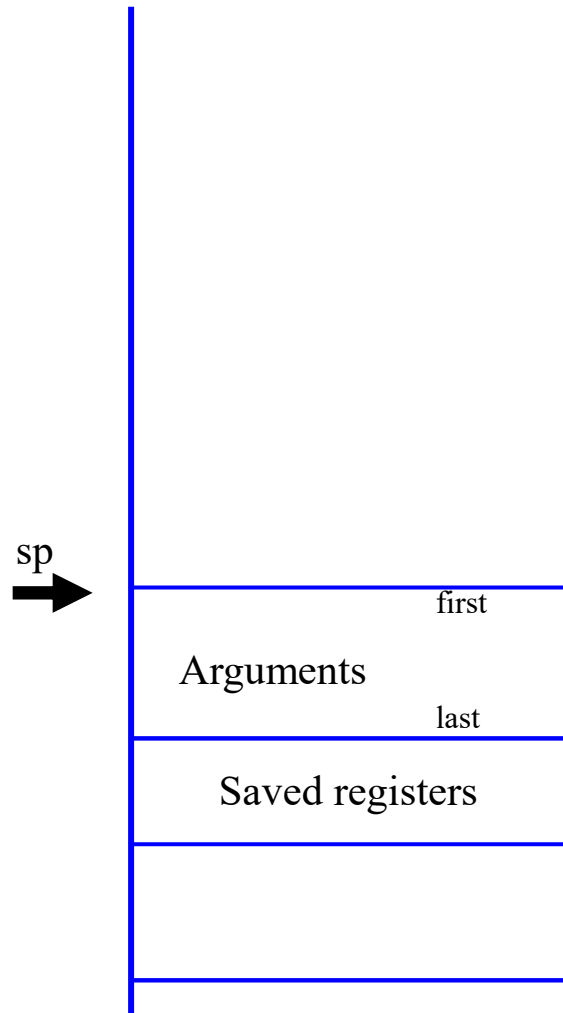
Construction of the stack frame caller function

Example (10 arguments):

Argument passing:

Before calling the calling
procedure:

- Leave the **first eight arguments in a_x (f_x)**
- The **rest of the arguments goes to the stack**



Stack frame
of the function
that makes the call

```
li  a0, 1
li  a1, 2
li  a2, 3
li  a3, 4
li  a4, 5
li  a5, 6
li  a6, 7
li  a7, 8
```

```
addi sp, sp, -8
```

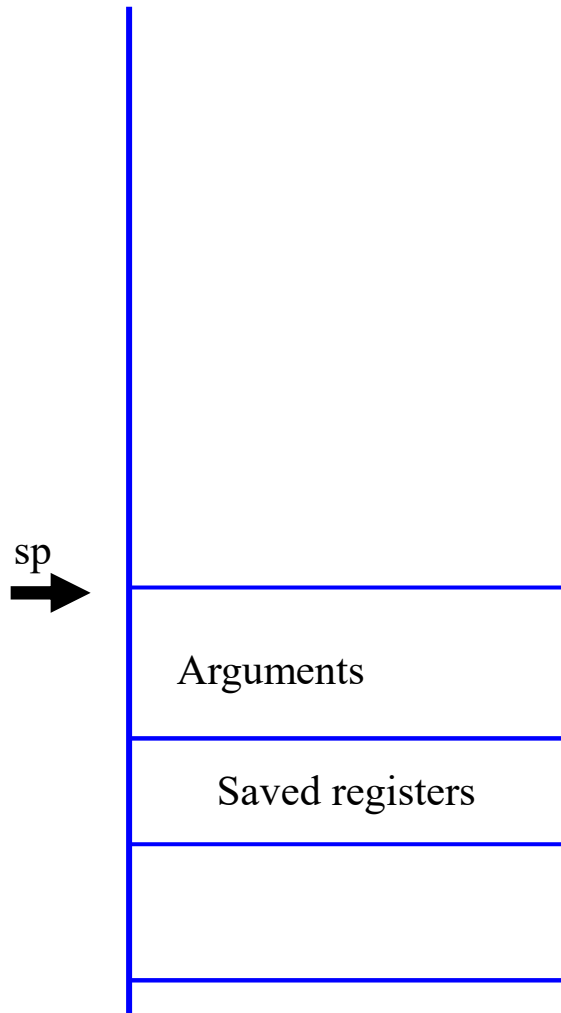
```
li  t0, 10
sw  t0, 4(sp)
```

```
li  t0, 9
sw  t0, 0(sp)
```

Construction of the stack frame caller function

Function invocation:

jal ra subrutina



li a0, 1
li a1, 2
li a2, 3
li a3, 4
li a4, 5
li a5, 6
li a6, 7
li a7, 8

addi sp, sp, -8

li t0, 10
sw t0, 4(sp)

li t0, 9
sw t0, 0(sp)

jal ra, subrutina

Construction of the stack frame called function

Stack frame allocation:

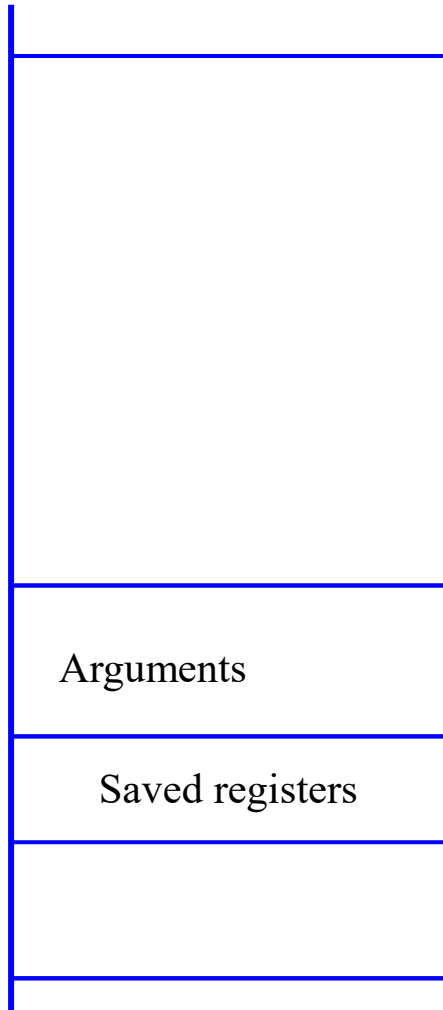
$sp = sp - \langle \text{frame size} \rangle$

Espacio para:

- ra
- s0...s7
- Local variables

Stack frame
of the function
that makes the call

sp
→



Example:

function:

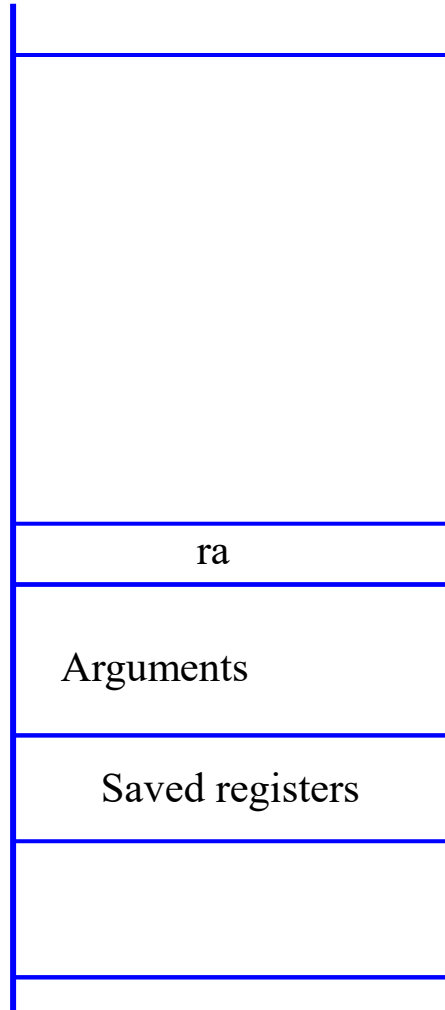
subu sp, sp, <fr.sz.>

Construction of the stack frame called function

Save registers that we allocated space for:

- ra (return address)

sp
→



ra is stored in non-terminal functions

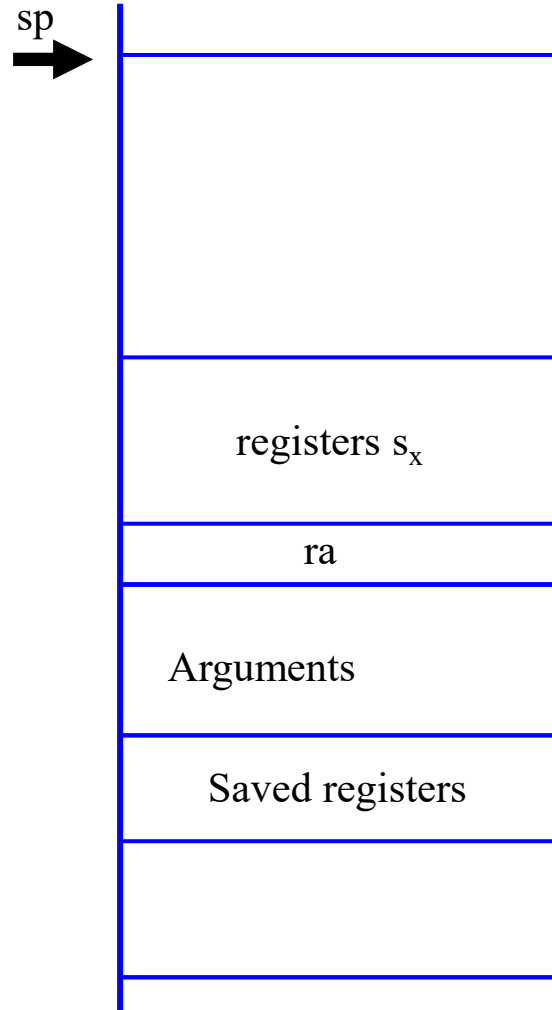
Stack frame
of the function
that makes the call

Construction of the stack frame called function

Save registers that we allocated space for:

- The s_x registers to be modified must be saved.
- A function cannot, by convention, modify the s_x registers (the t_x and the a_x can be modified).

Stack frame
of the function
that makes the call



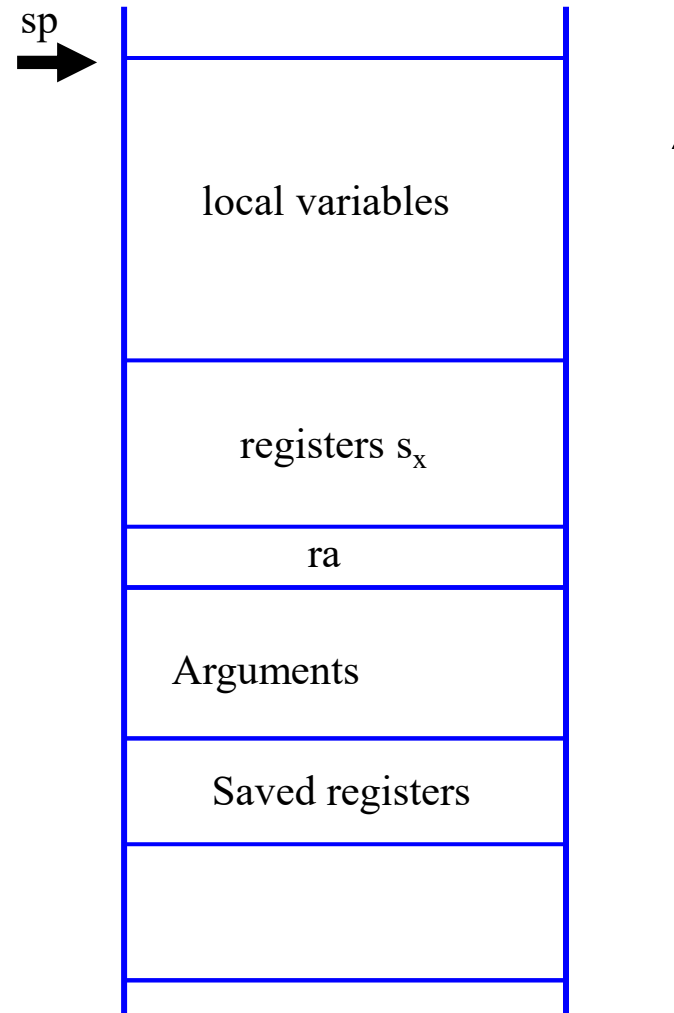
Example:

function:

```
subu sp, sp, <fr.sz.>
sw ra, <fr.sz-4>(sp)
sw s0, <fr.sz-8>(sp)
sw sl,    <...>(sp)
...
```

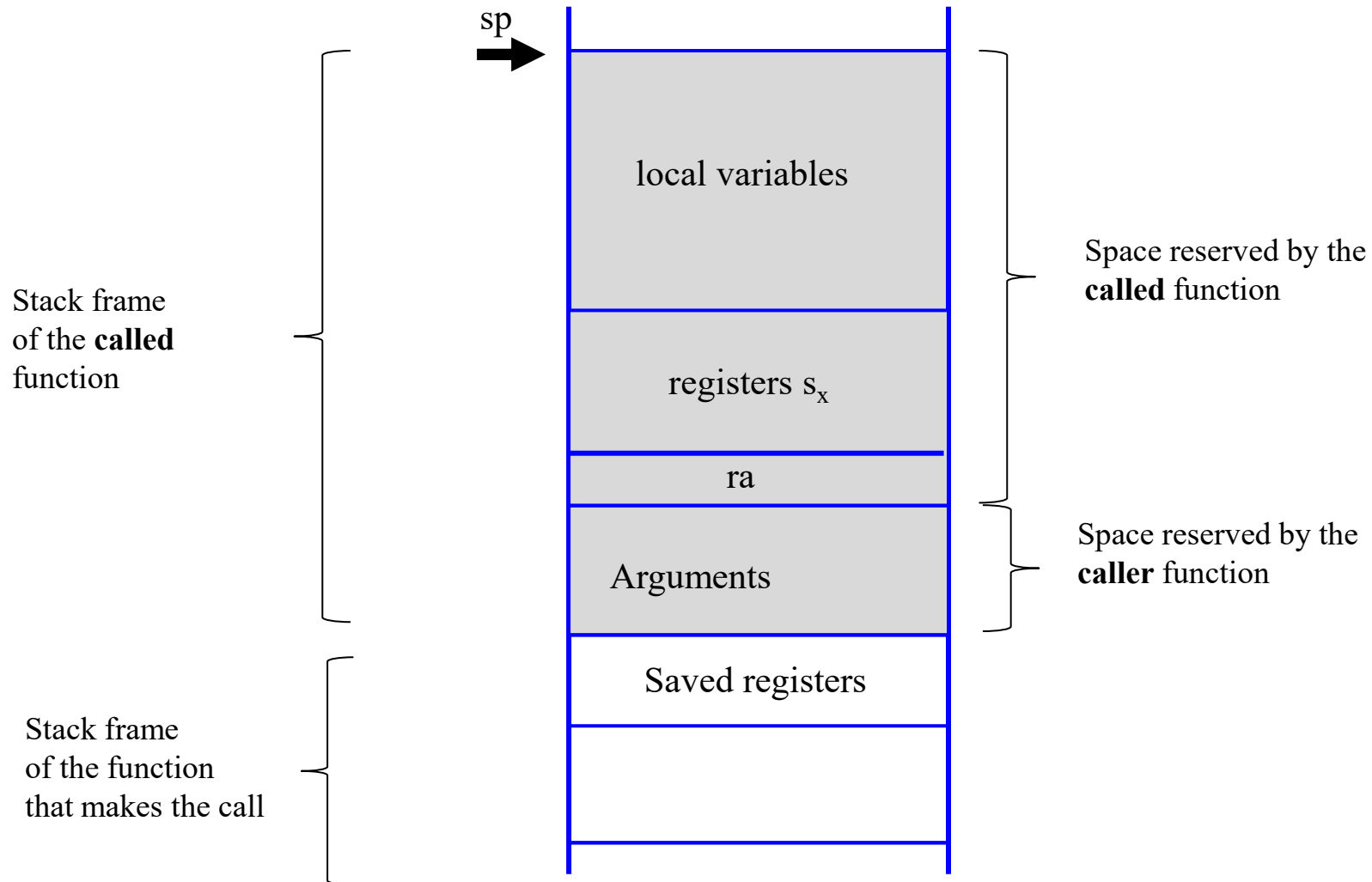
Construction of the stack frame called function

- Space for local variables



Stack frame
of the function
that makes the call

Stack frame construction



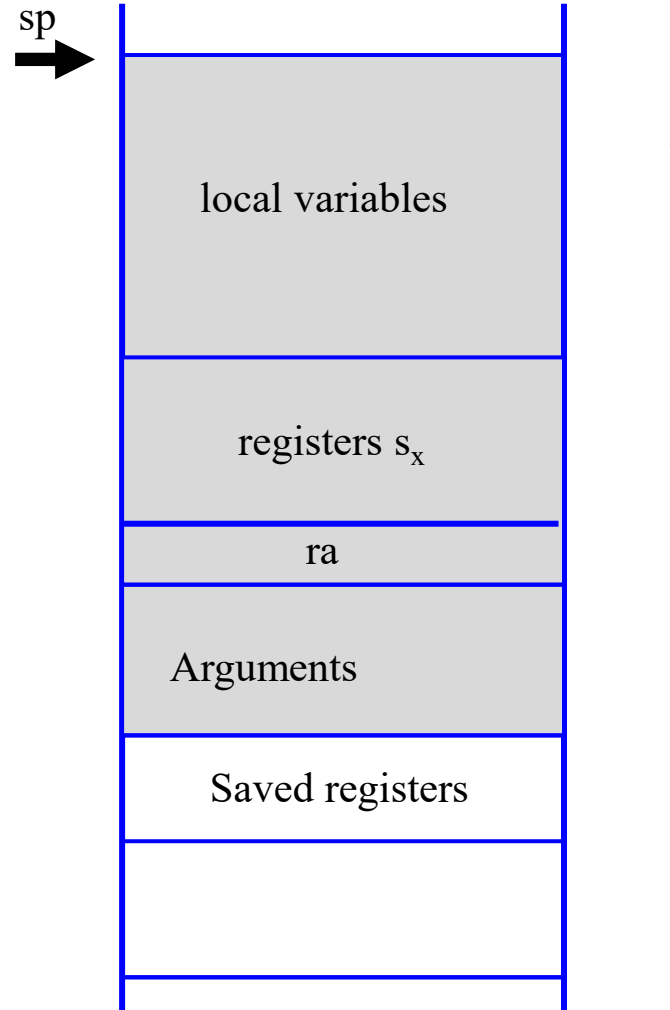
Subroutine termination called **d** function

The results are returned:

Use the appropriated registers:
a0, a1, (fa0, fa1)

If more complex structures
need to be returned, they are
left on the stack (the caller
must allocate the space)

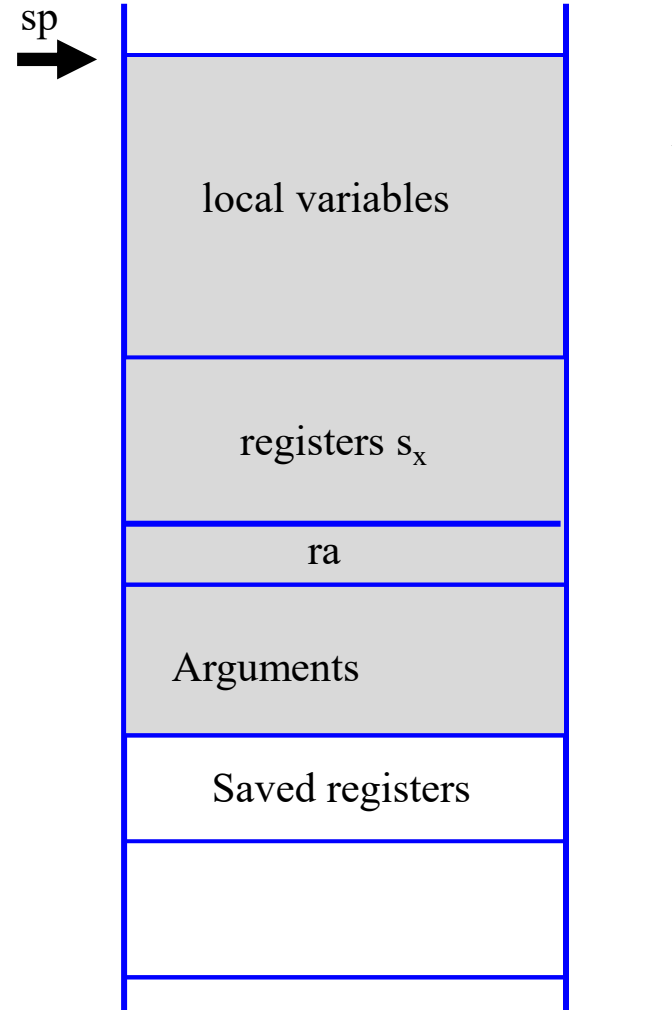
Stack frame
of the function
that makes the call



Subroutine termination called **d** function

**The saved register are
restored:**

registers s_x
register ra

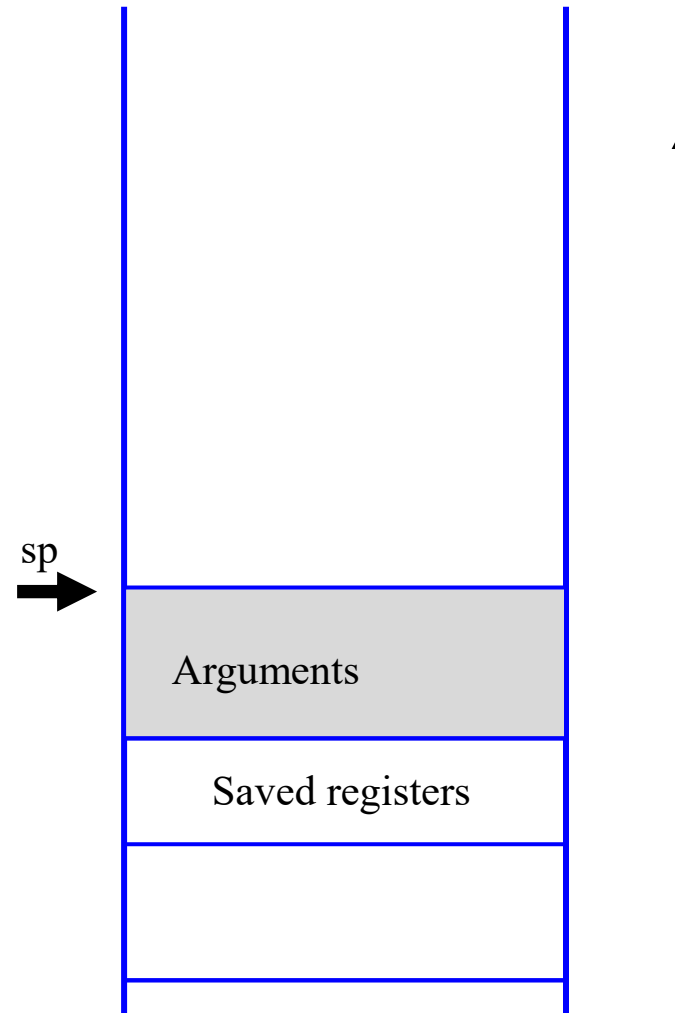


Stack frame
of the function
that makes the call

Subroutine termination called **d** function

Stack frame is freed:

$sp = sp + \text{<frame size>}$

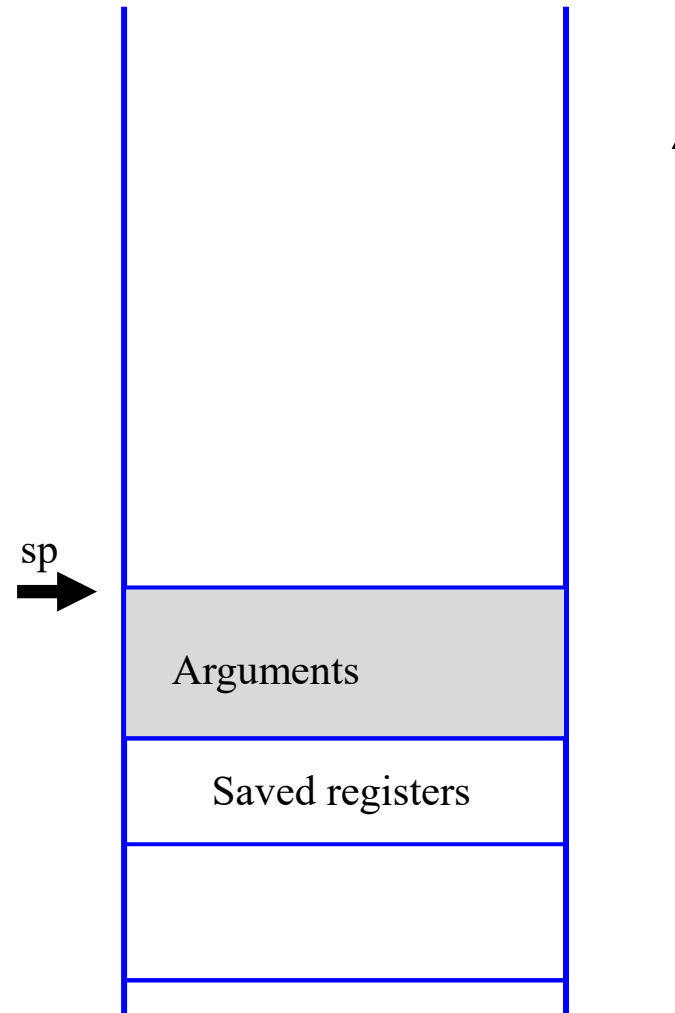


Stack frame
of the function
that makes the call

Subroutine termination called **d** function

Function returns:

`jr ra`



Stack frame
of the function
that makes the call

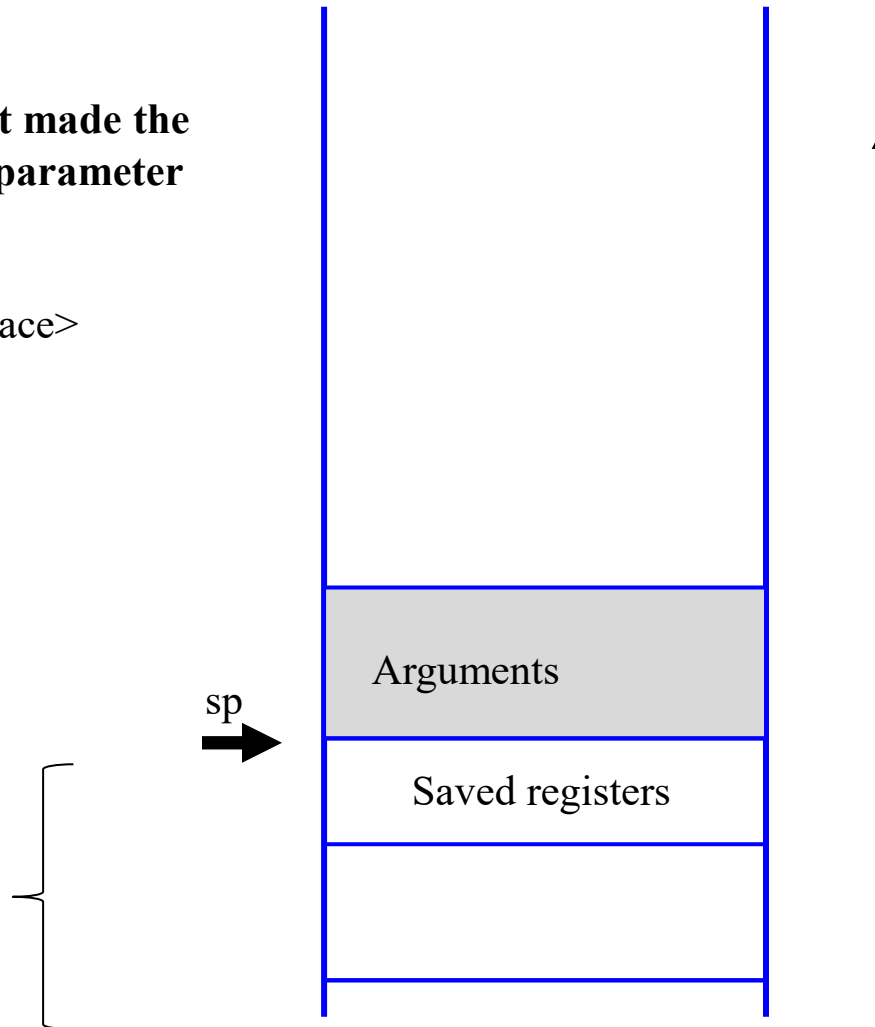
Subroutine termination

calle function

The function that made the call frees up the parameter space:

$sp = sp + \langle \text{arg. space} \rangle$

Stack frame
of the function
that makes the call



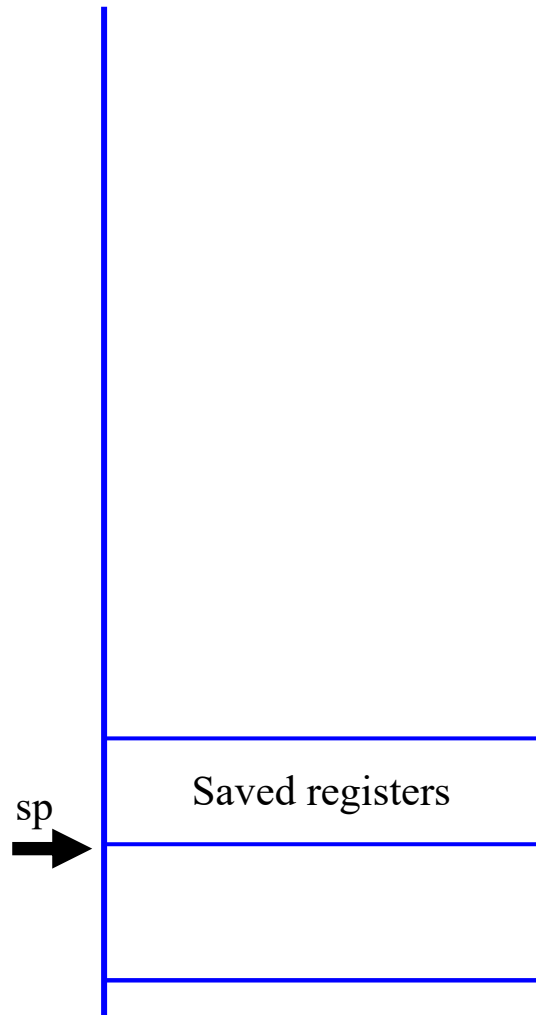
Subroutine termination

call function

The function that made the call restores the registers it saved.

Restore the `sp` register to its initial position

Stack frame of the function that makes the call



Example:

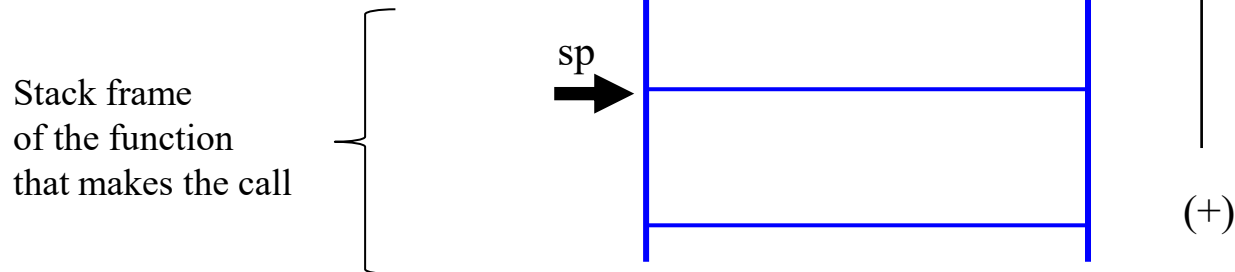
```
addi sp, sp, -8  
sw  t0, 0(sp)  
sw  t1, 4(sp)
```

```
li  a0, 5  
jal ra, funcion
```

```
lw  t0, 0(sp)  
lw  t1, 4(sp)  
add sp, sp, 8
```

State after subroutine termination

- **Initial** situation before calling a function



Local variables in registers

- ▶ Whenever possible, local variables (int, double, char, ...) are stored in registers
 - ▶ If registers cannot be used (there are not enough), the stack is used

```
int f(...)  
{  
    int i, j, k;  
  
    i = 0;  
    j = 1;  
    k = i + j;  
    . . .  
}
```

```
f:    . . .  
      li    t0, 0  
      li    t1, 1  
      add   t2, t0, t1  
      . . .
```

ARCOS Group

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L3: Fundamentals of assembler programming (4) Computer Structure

Bachelor in Computer Science and Engineering

Bachelor in Applied Mathematics and Computing

Dual Bachelor in Computer Science and Engineering and Business Administration

